

CSA13 – CLOUD COMPUTING AND BIG DATA ANALYTICS

LAB RECORD – LIST OF EXPERIMENTS

Ex. No: 12 Demonstration of Virtualization Using VMware Workstation and Creation of a Virtual Machine

Aim

Demonstrate virtualization by installing a Type-2 Hypervisor and creating/configuring a VM with a host operating system.

PROCEDURE

Step 1: Install VMware Workstation on the host computer.

Step 2: Open VMware Workstation.

Step 3: Click Create a New Virtual Machine.

Step 4: Select Typical (Recommended) configuration.

Step 5: Select the required Windows/Linux ISO image.

Step 6: Enter the Virtual Machine name and select the storage location.

Step 7: Configure the processor according to the given requirement.

Step 8: Allocate the required RAM to the Virtual Machine.

Step 9: Create a virtual hard disk with the required storage capacity.

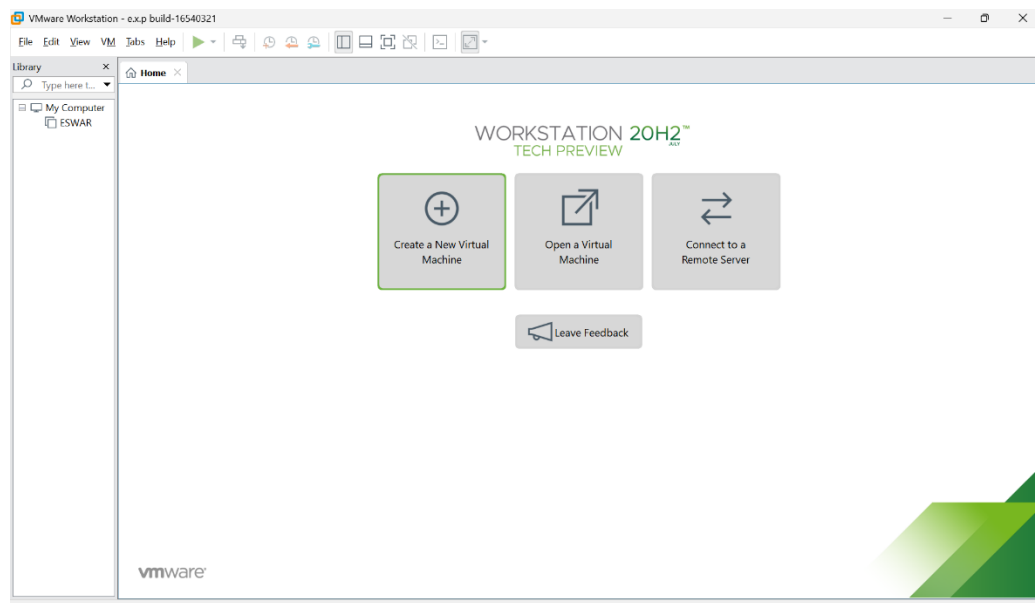
Step 10: Click Finish to create the Virtual Machine.

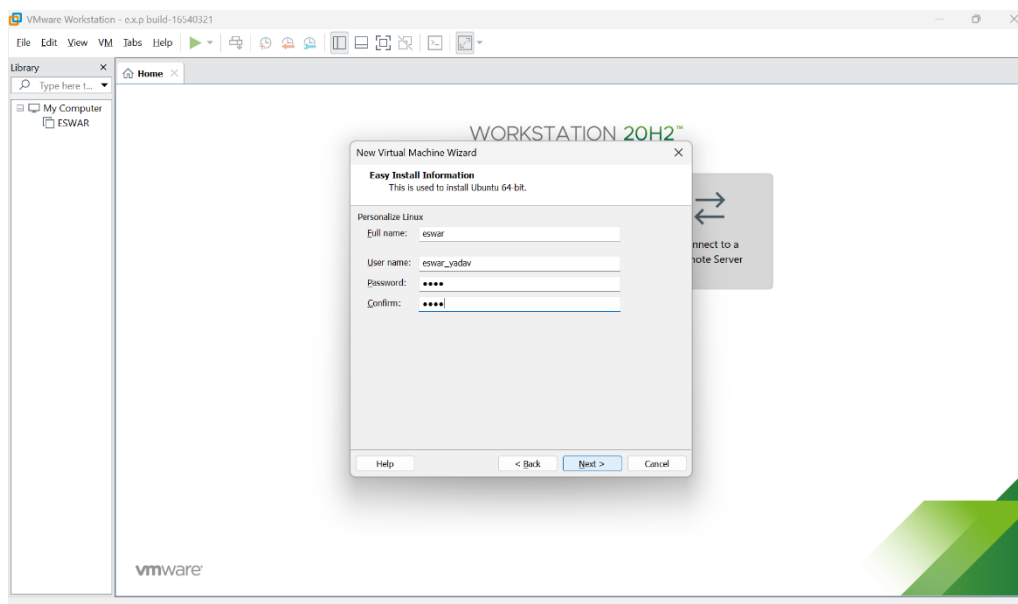
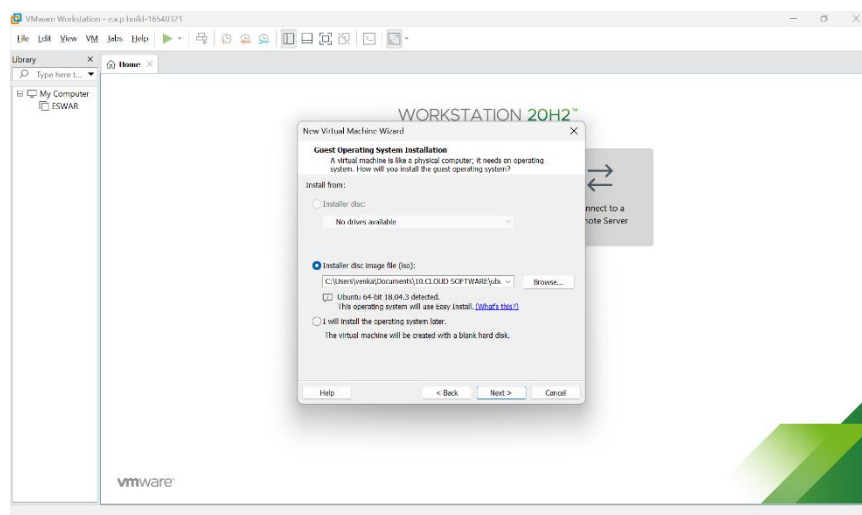
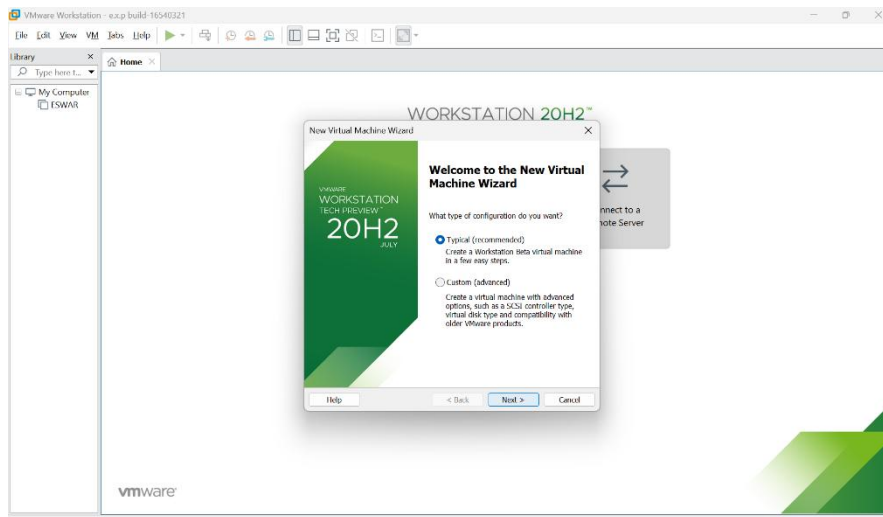
Step 11: Power on the Virtual Machine and install the selected operating system.

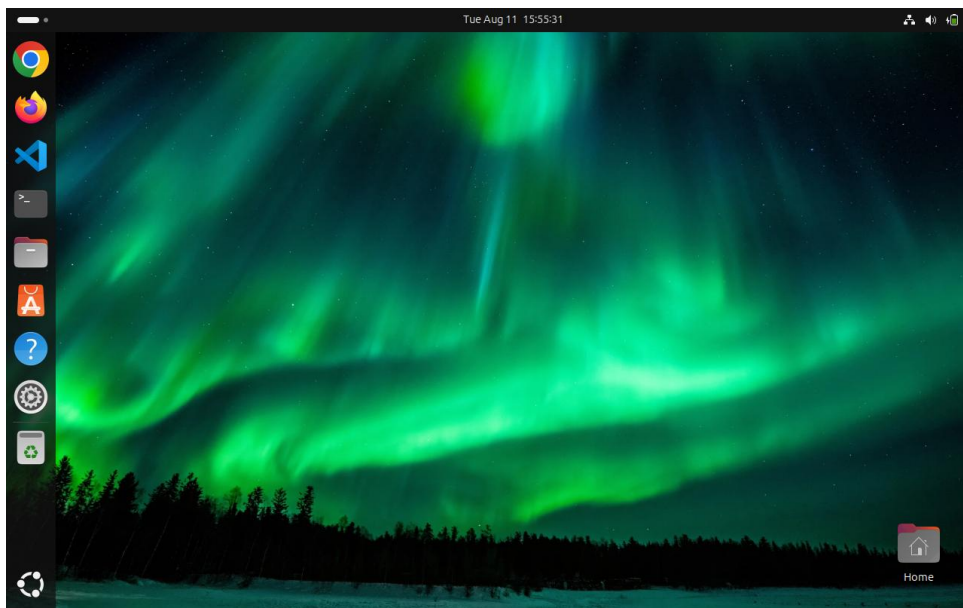
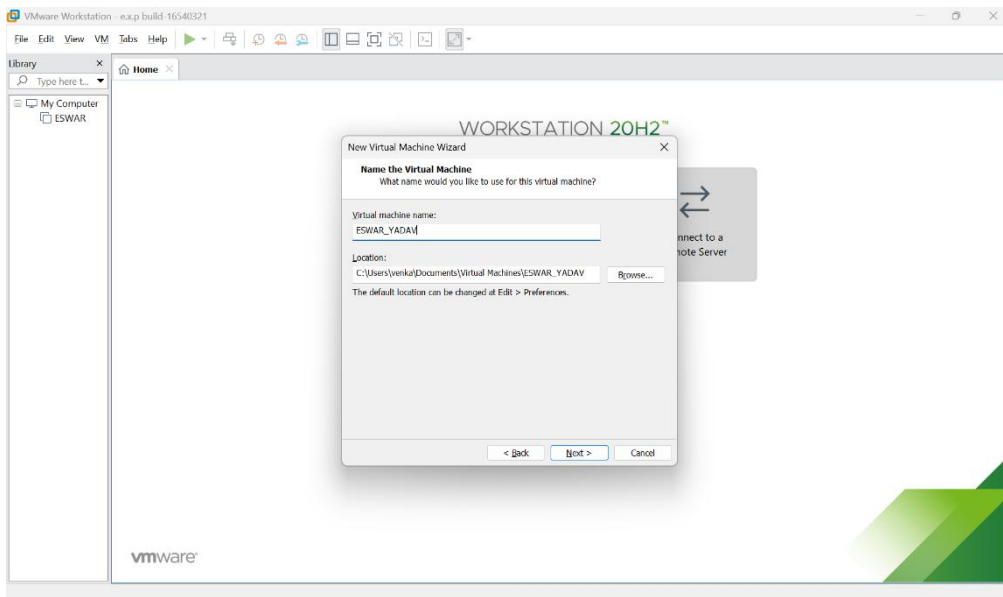
Step 12: Verify the CPU, RAM, and storage configuration.

Step 13: The Virtual Machine has been successfully created using VMware Workstation.

Output







Result

Thus the Virtual Machine was successfully created using **VMware Workstation**, a Type-2 Hypervisor, with the specified CPU, RAM, and storage configuration.

Ex. No: 13 Create a Virtual Machine with 1 CPU, 2 GB RAM and 15 GB Storage Using VMware Workstation

Aim

To create and configure a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage using VMware Workstation.

PROCEDURE

Step 1: Open VMware Workstation.

Step 2: Click Create a New Virtual Machine.

Step 3: Select Typical (Recommended).

Step 4: Select the required Windows/Linux ISO.

Step 5: Enter the VM name and storage location.

Step 6: Set the Processor to 1 CPU.

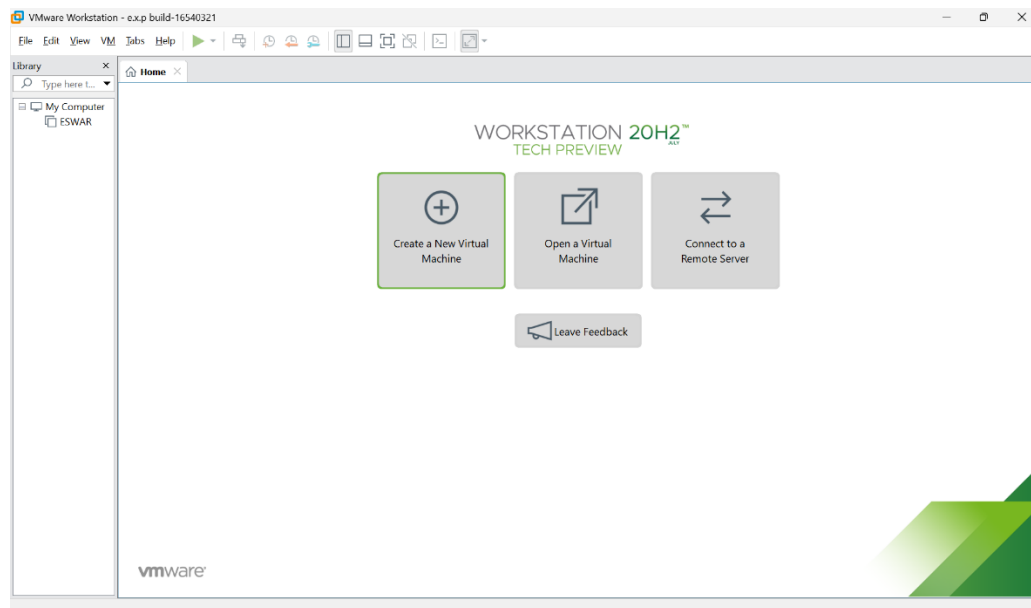
Step 7: Allocate 2 GB RAM.

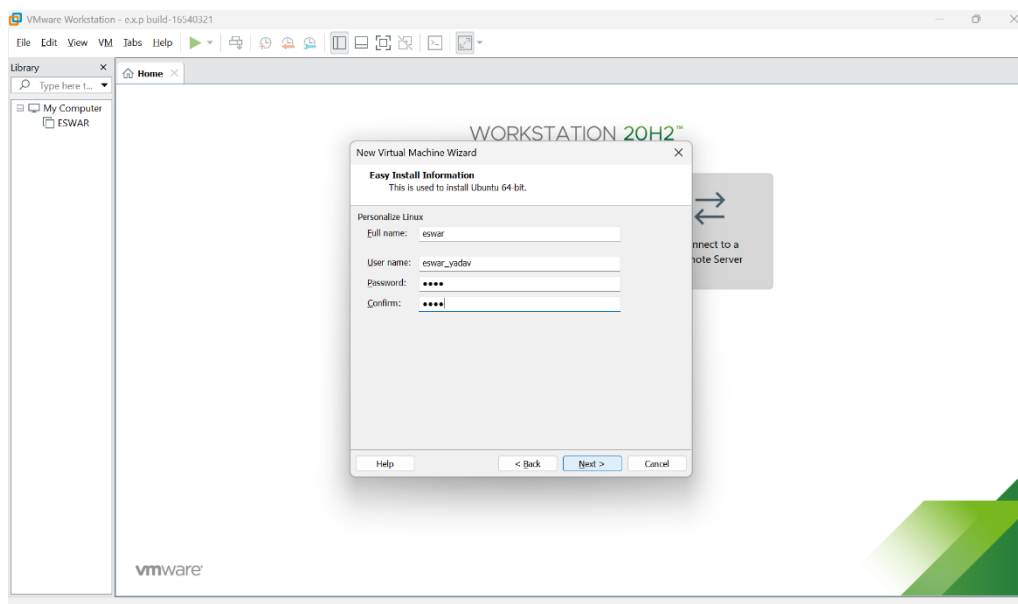
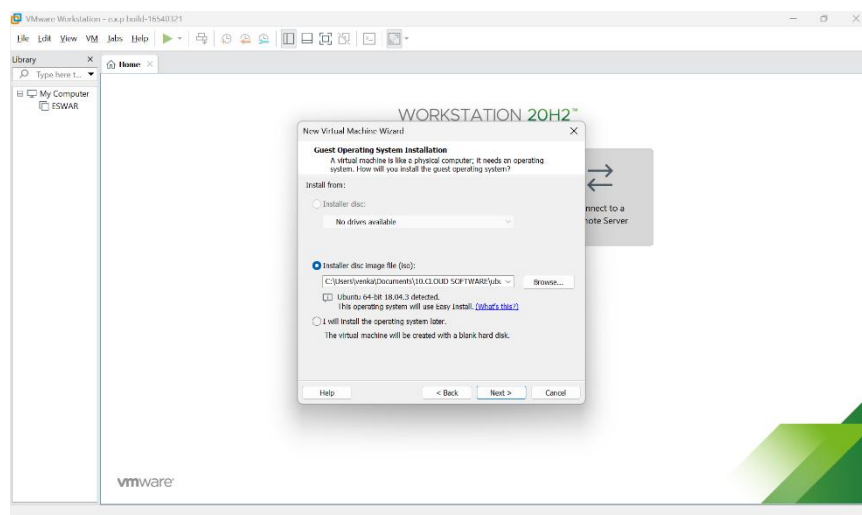
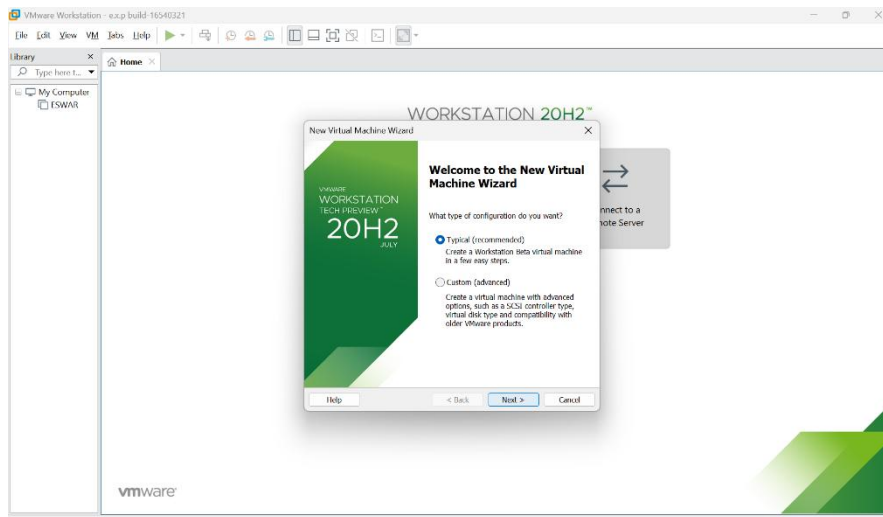
Step 8: Create a 15 GB virtual hard disk.

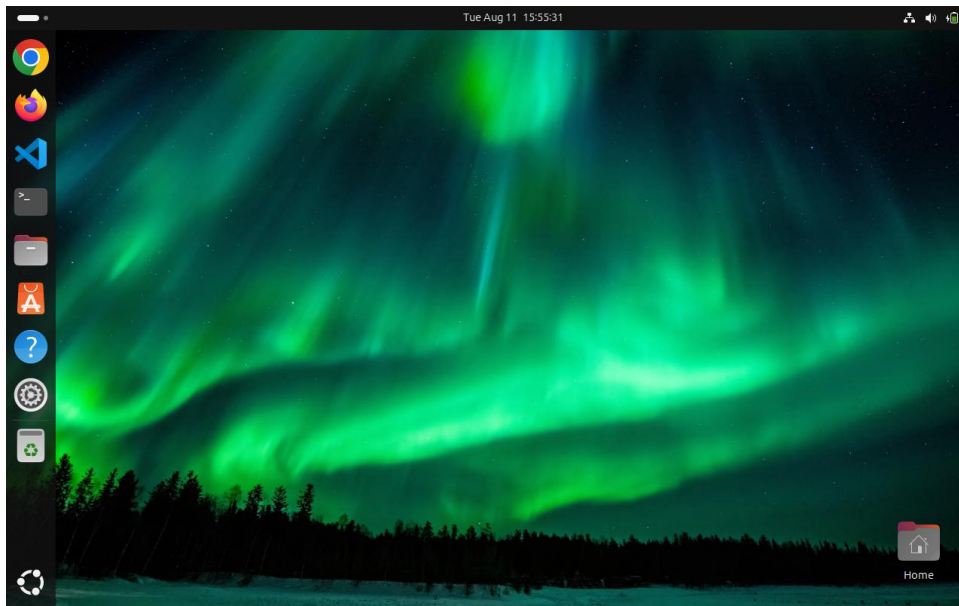
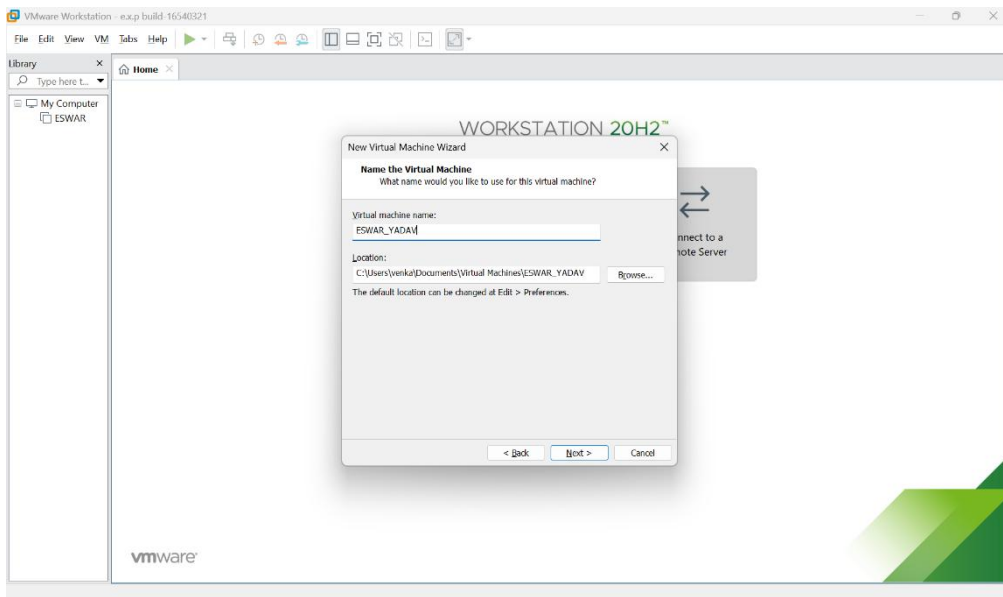
Step 9: Click Finish to create the VM.

Step 10: Power on the VM and verify the configuration.

Output







Result

The virtual machine was successfully created and configured with 1 CPU, 2 GB RAM, and 15 GB storage using VMware Workstation.

Ex. No: 14 Create a Virtual Hard Disk and Allocate Storage Using VMware Workstation

Aim

To create a Virtual Hard Disk and allocate storage to a virtual machine using VMware Workstation.

PROCEDURE

Step 1: Open VMware Workstation.

Step 2: Select the required virtual machine.

Step 3: Click Edit Virtual Machine Settings.

Step 4: Select Hard Disk and click Add.

Step 5: Select Hard Disk as the hardware type.

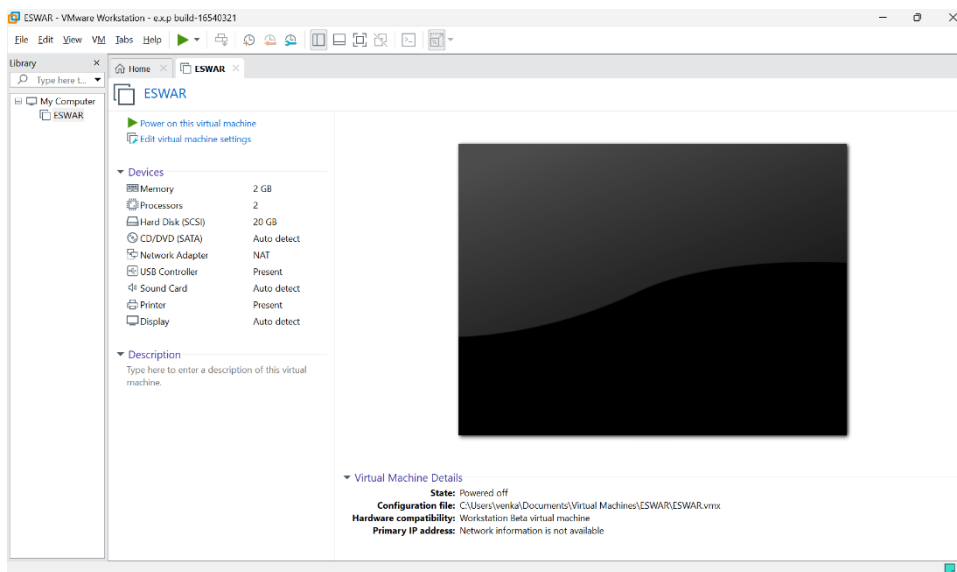
Step 6: Select Create a new virtual disk.

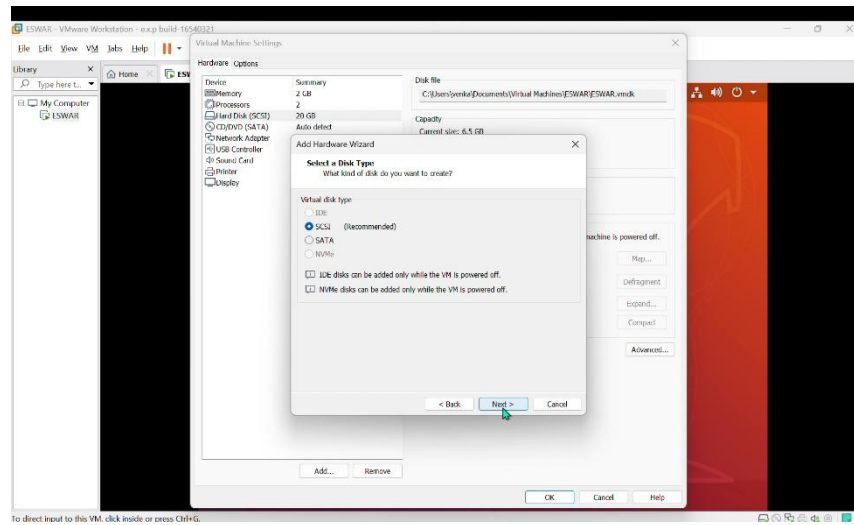
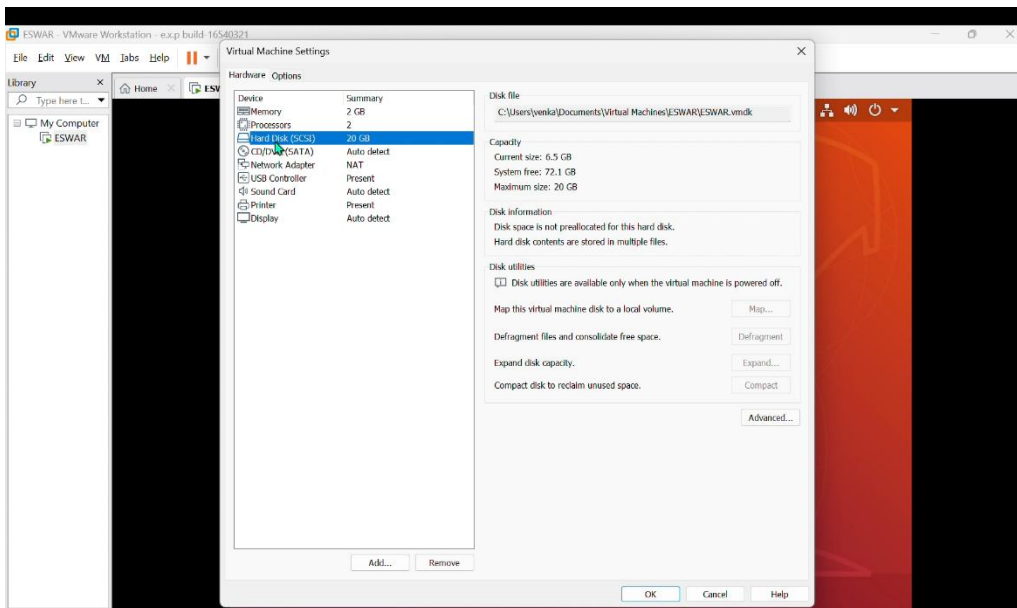
Step 7: Select the required disk size and storage location.

Step 8: Click Finish to create the virtual hard disk.

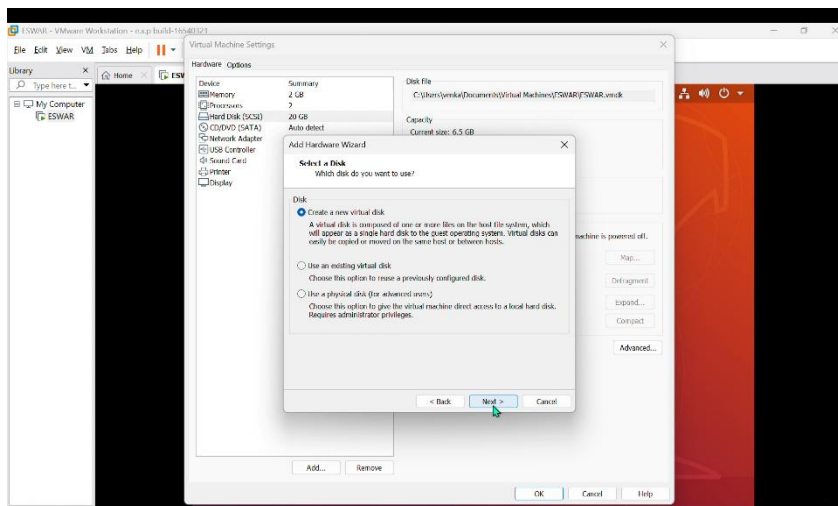
Step 9: Click OK and start the virtual machine.

Output

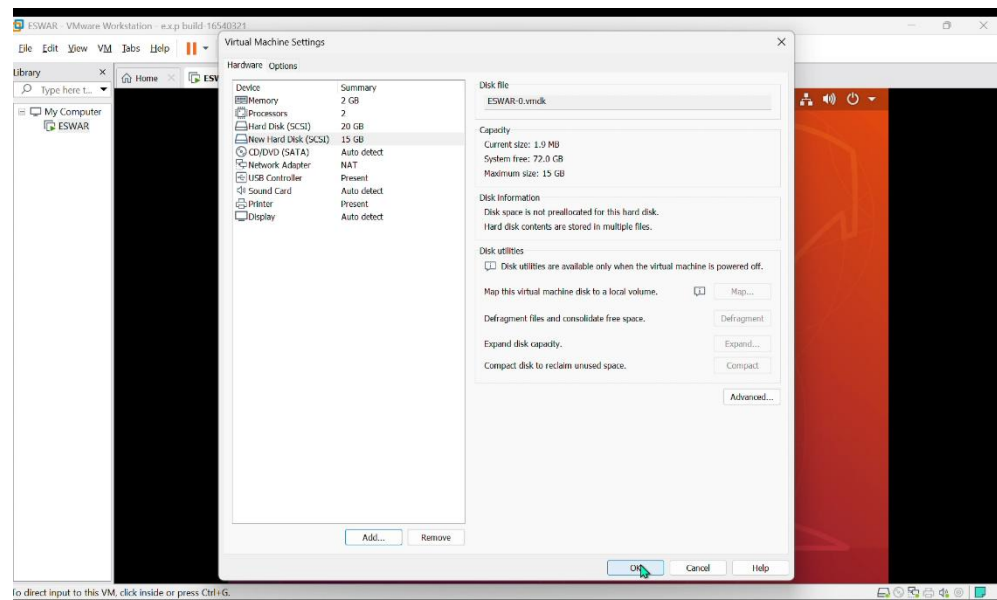
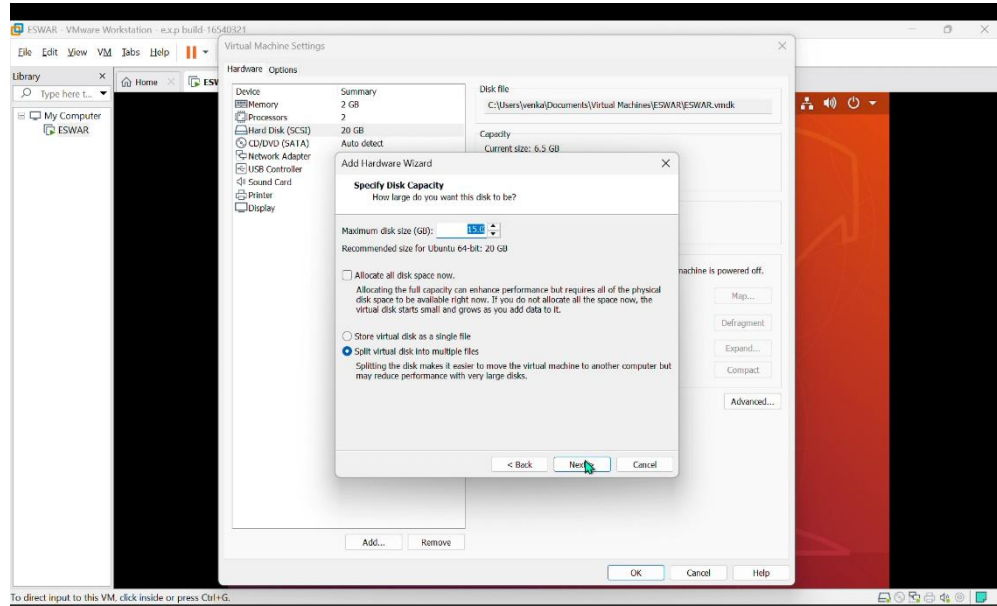




To direct input to this VM, click inside or press Ctrl+G.



To direct input to this VM, click inside or press Ctrl+G.



Result

Thus the Virtual Machine was successfully created using **VMware Workstation**, a Type-2 Hypervisor, with the specified CPU, RAM, and storage configuration.

Ex. No: 15 Creating and Testing a VM Snapshot

Aim

To demonstrate virtualization by creating a Snapshot of a Virtual Machine using VMware Workstation and testing it by loading the previous version of the Virtual Machine.

PROCEDURE

Step 1: Open VMware Workstation.

Step 2: Select the previously created Virtual Machine.

Step 3: Power on the Virtual Machine.

Step 4: Make some changes inside the Virtual Machine.

Step 5: Select VM → Snapshot → Take Snapshot.

Step 6: Enter a suitable snapshot name.

Step 7: Click Take Snapshot to save the current state of the Virtual Machine.

Step 8: Make some additional changes to the Virtual Machine.

Step 9: Select VM → Snapshot → Snapshot Manager.

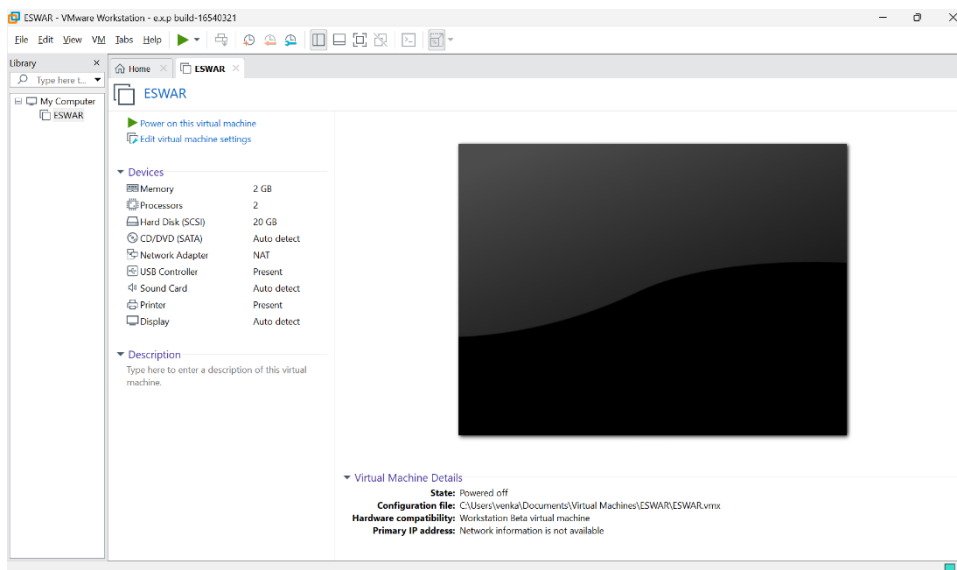
Step 10: Select the previously created snapshot.

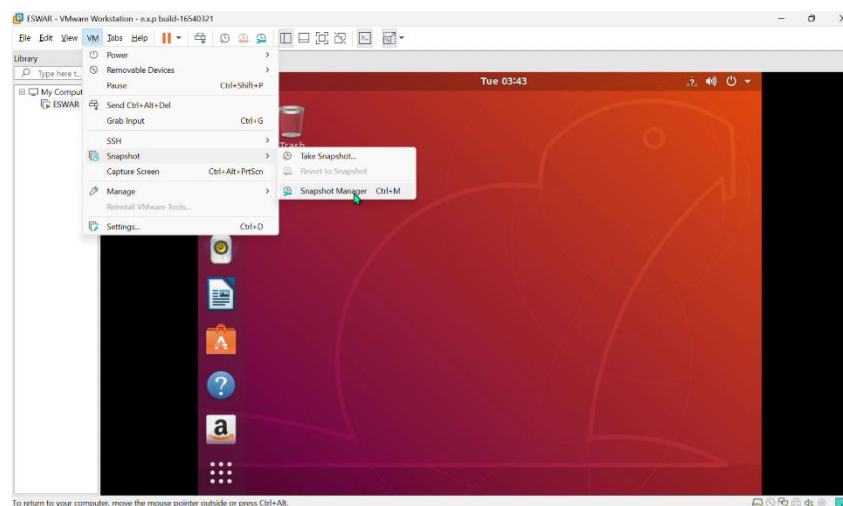
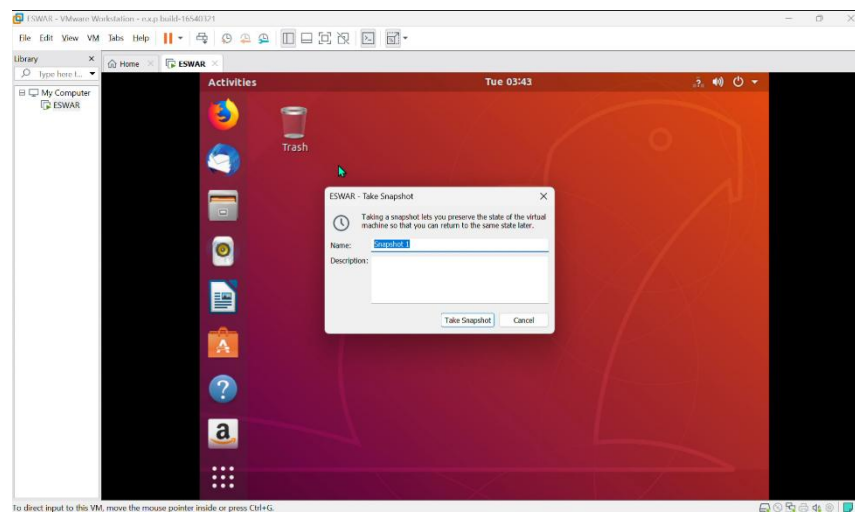
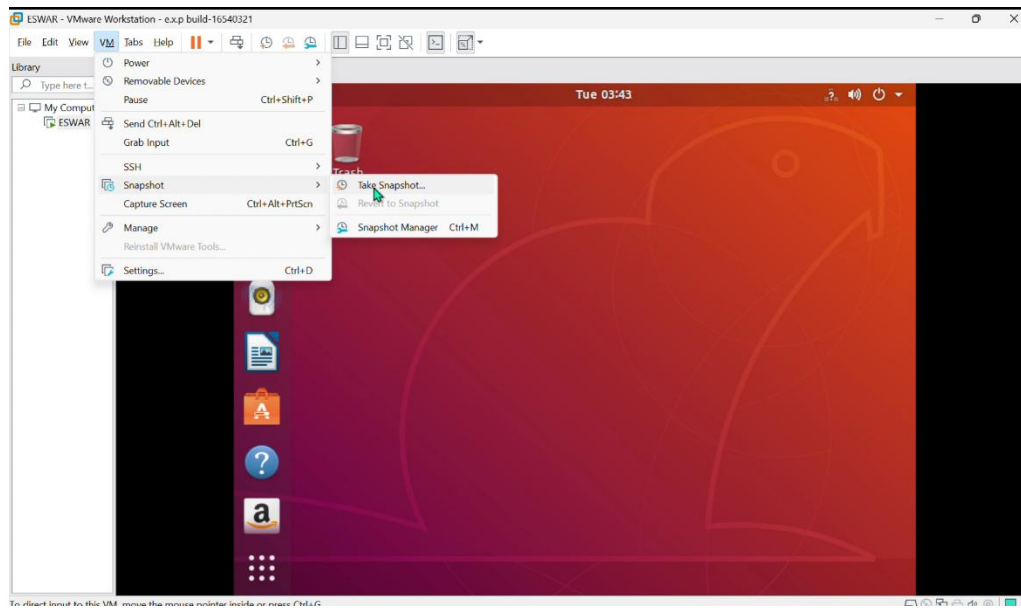
Step 11: Click Go To to restore the Virtual Machine to the saved snapshot.

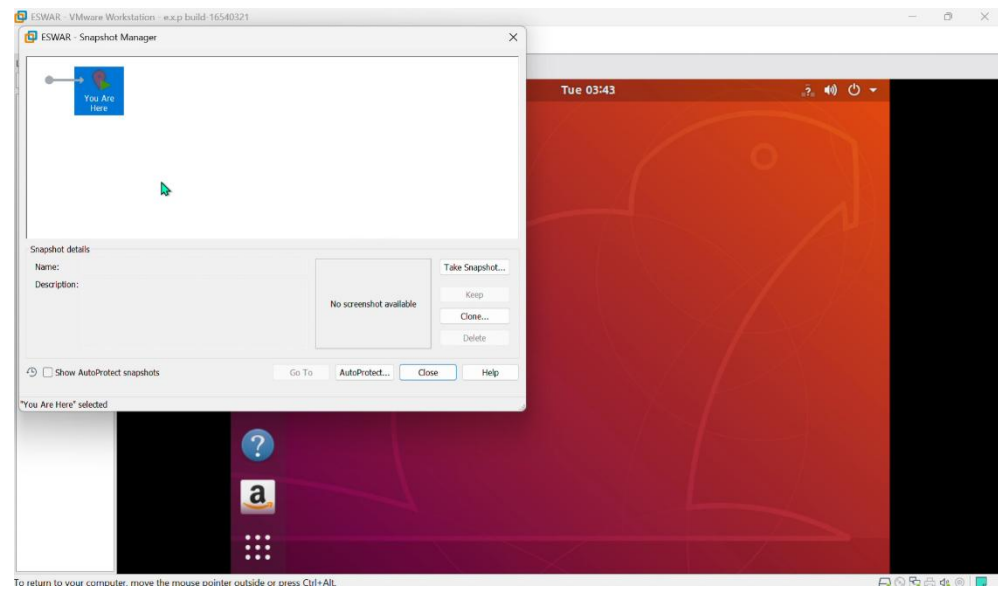
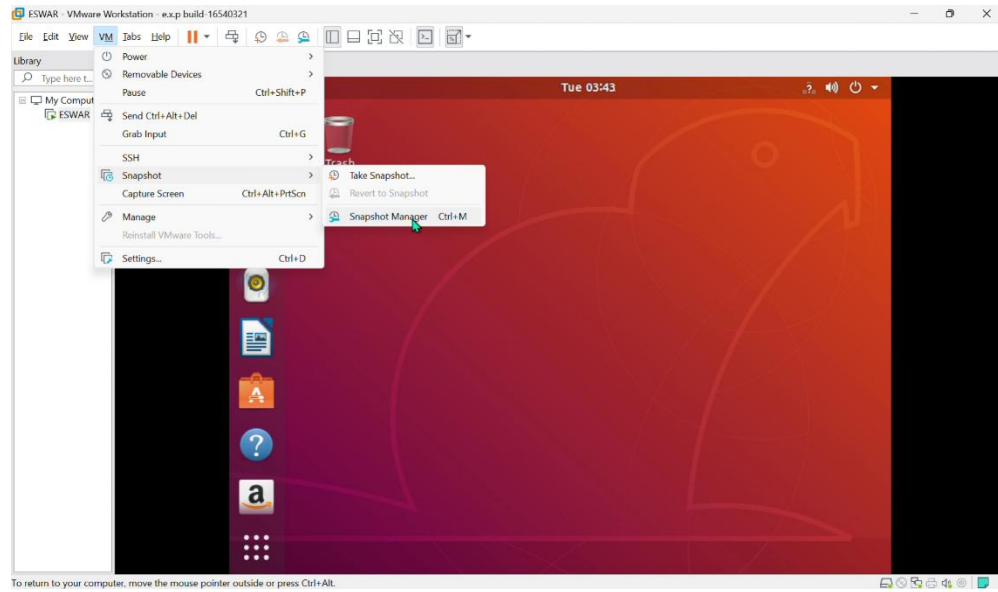
Step 12: Verify that the Virtual Machine has returned to its previous state.

Step 13: The Snapshot has been successfully created and tested.

Output







Result

The VM snapshot was successfully created, and the virtual machine was restored to its previous state using **VMware Workstation**.

Ex. No: 16 Create and Test a Clone of a Virtual Machine Using VMware Workstation

Aim

To create a clone of a virtual machine and test the cloned VM using VMware Workstation.

PROCEDURE

Step 1: Open VMware Workstation and select the virtual machine.

Step 2: Power off the virtual machine.

Step 3: Select VM → Manage → Clone.

Step 4: Select the current state of the virtual machine.

Step 5: Choose Create a Full Clone.

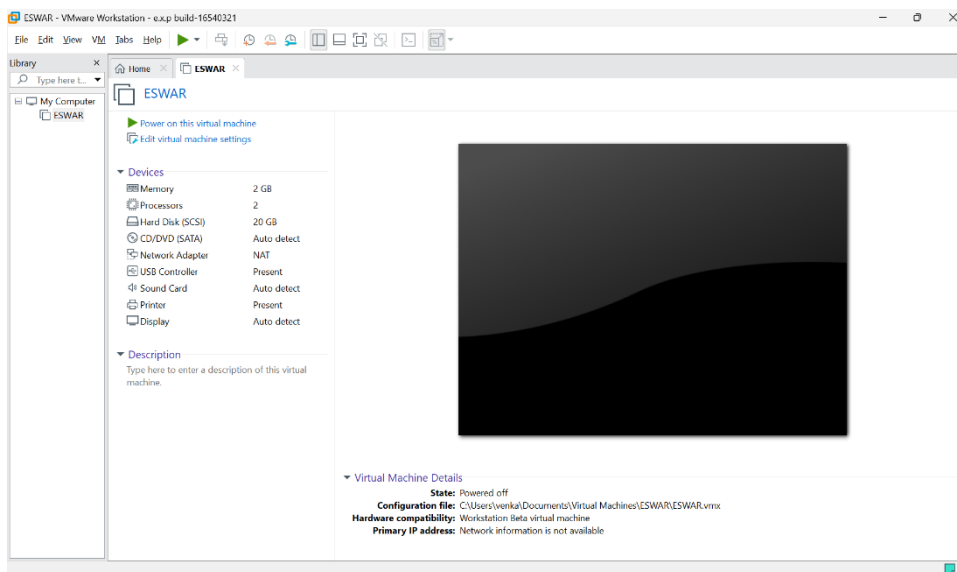
Step 6: Enter the name and storage location for the cloned VM.

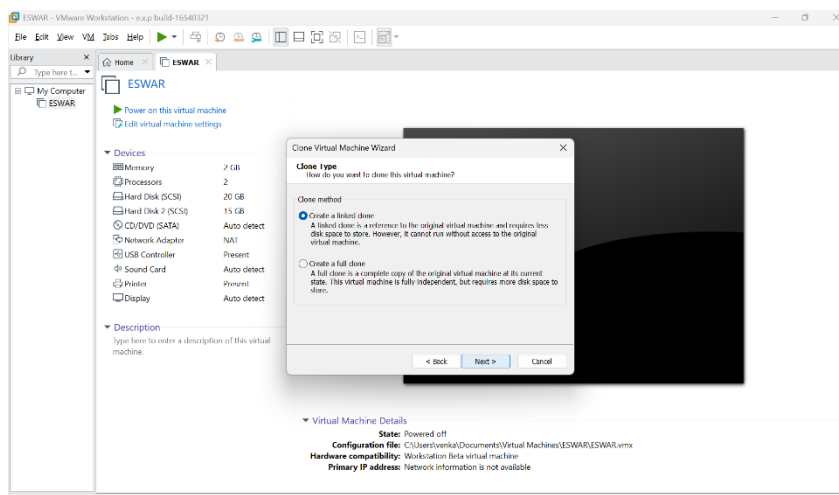
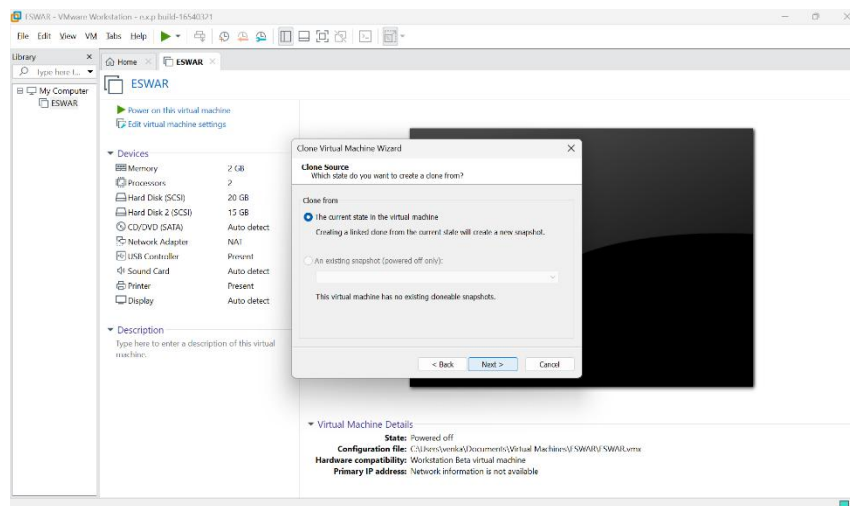
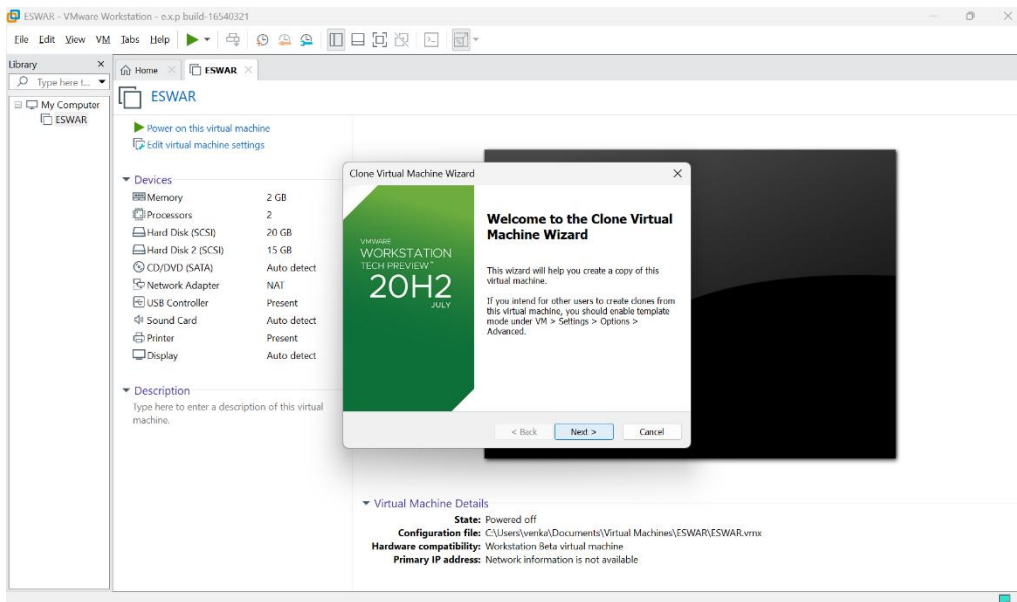
Step 7: Click Finish to create the clone.

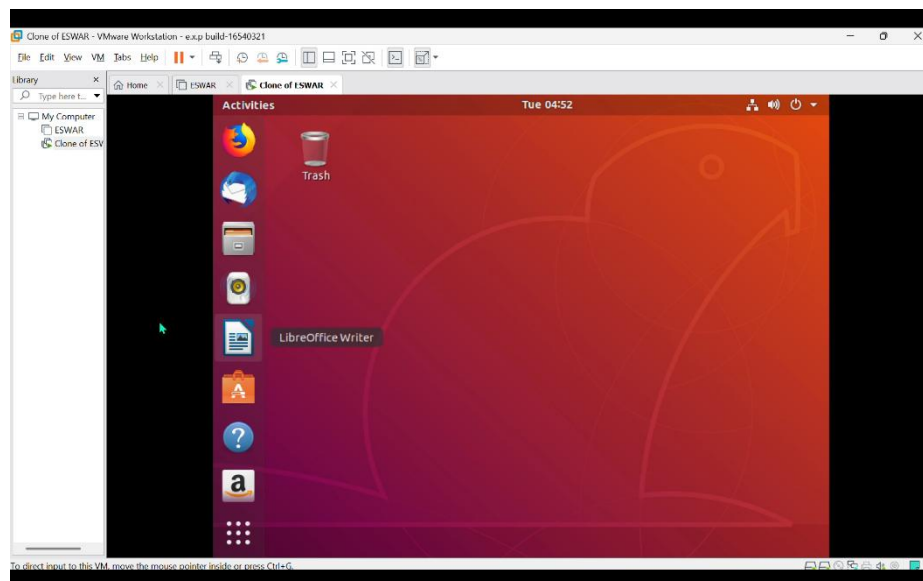
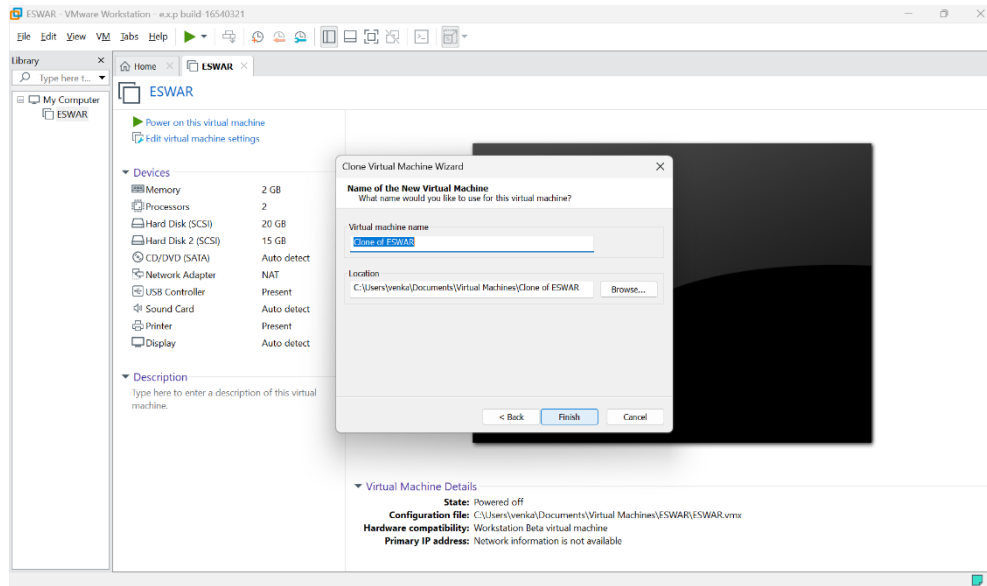
Step 8: Power on the cloned virtual machine.

Step 9: Verify that the cloned VM works correctly.

Output







Result

The virtual machine was successfully **cloned and tested** using VMware Workstation.

Ex. No: 17 Change Hardware Compatibility of a Virtual Machine and Run Unix Commands Using VMware Workstation

Aim

To change the hardware compatibility of an existing virtual machine and execute basic Unix commands using VMware Workstation.

PROCEDURE

Step 1: Open VMware Workstation and select the virtual machine.

Step 2: Power off the virtual machine.

Step 3: Select VM → Manage → Change Hardware Compatibility.

Step 4: Select the required hardware compatibility version.

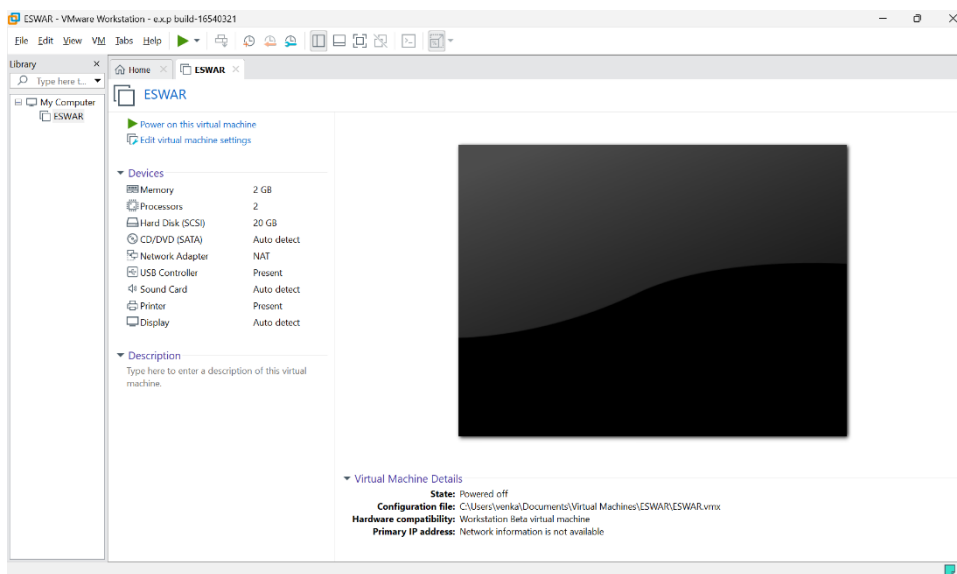
Step 5: Follow the wizard and create/update the compatible VM.

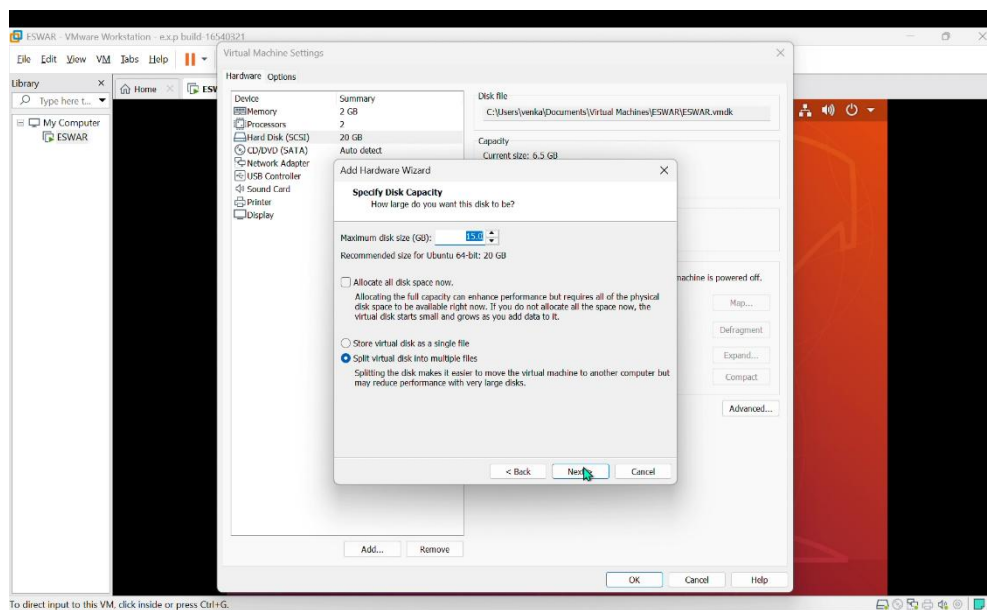
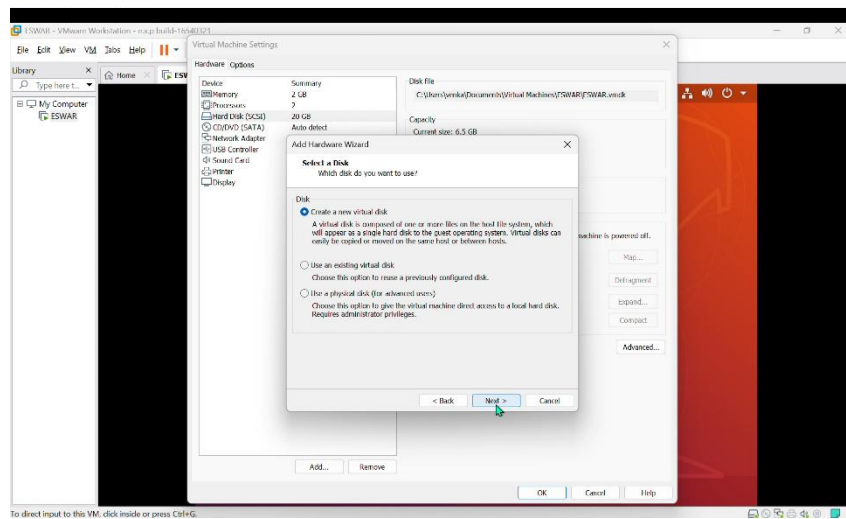
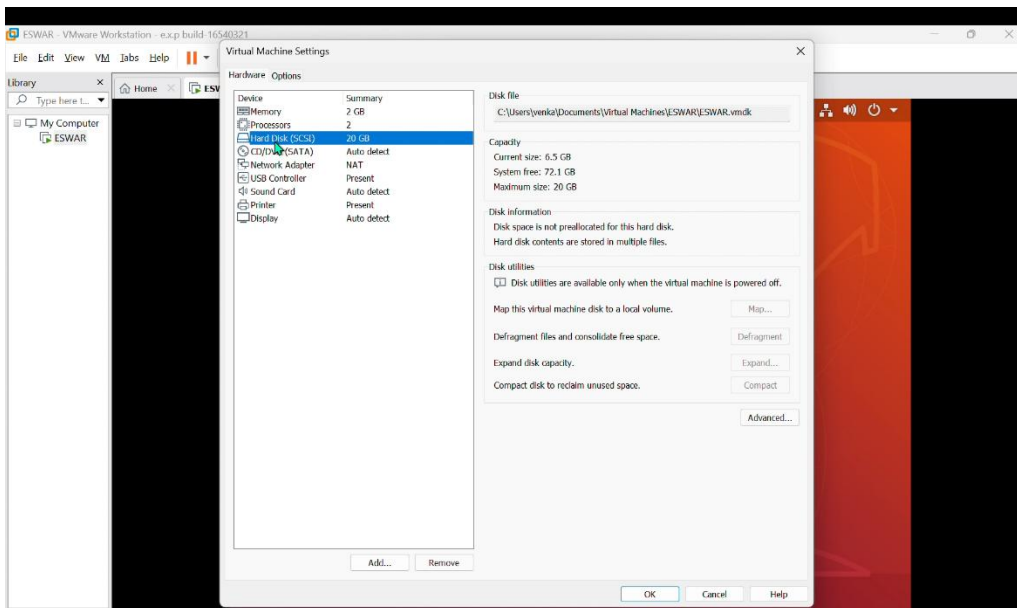
Step 6: Power on the virtual machine.

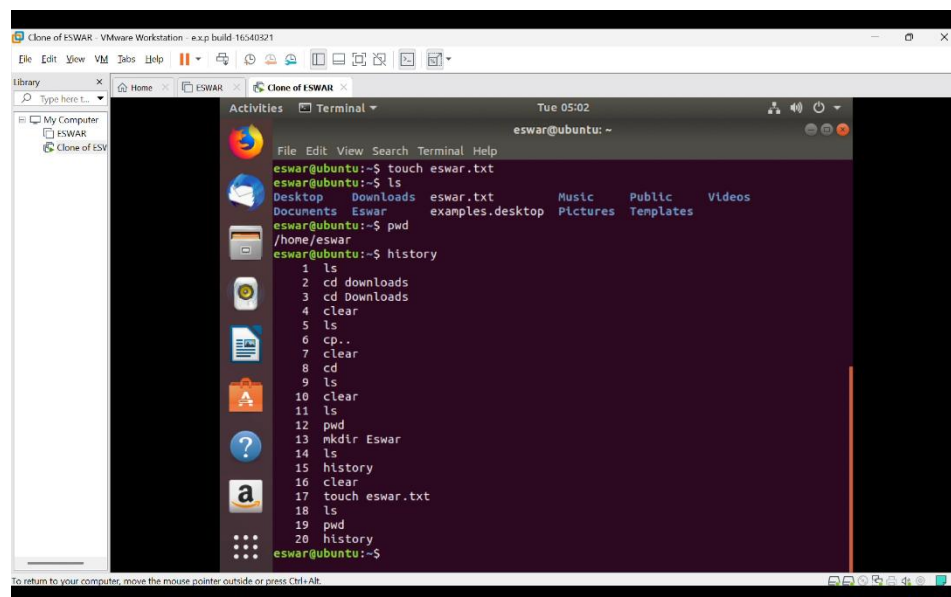
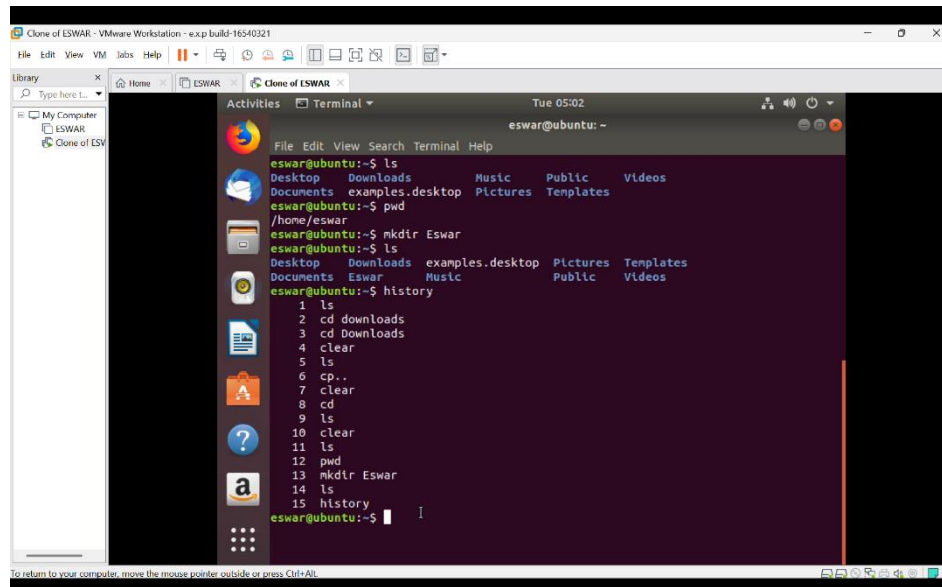
Step 7: Open the Unix/Linux terminal.

Step 8: Execute basic commands such as ls, pwd, cd, and mkdir.

Output







Result

The virtual machine was successfully **cloned and tested** using VMware Workstation.

Ex. No: 18 Install VMware Workstation with Linux/Windows OS on a Windows Host

Aim

To install VMware Workstation and run a Linux/Windows operating system as a virtual machine on a Windows host system.

PROCEDURE

Step 1: Install and open VMware Workstation on Windows.

Step 2: Click Create a New Virtual Machine.

Step 3: Select Typical (Recommended).

Step 4: Select the required Linux/Windows ISO.

Step 5: Enter the VM name and storage location.

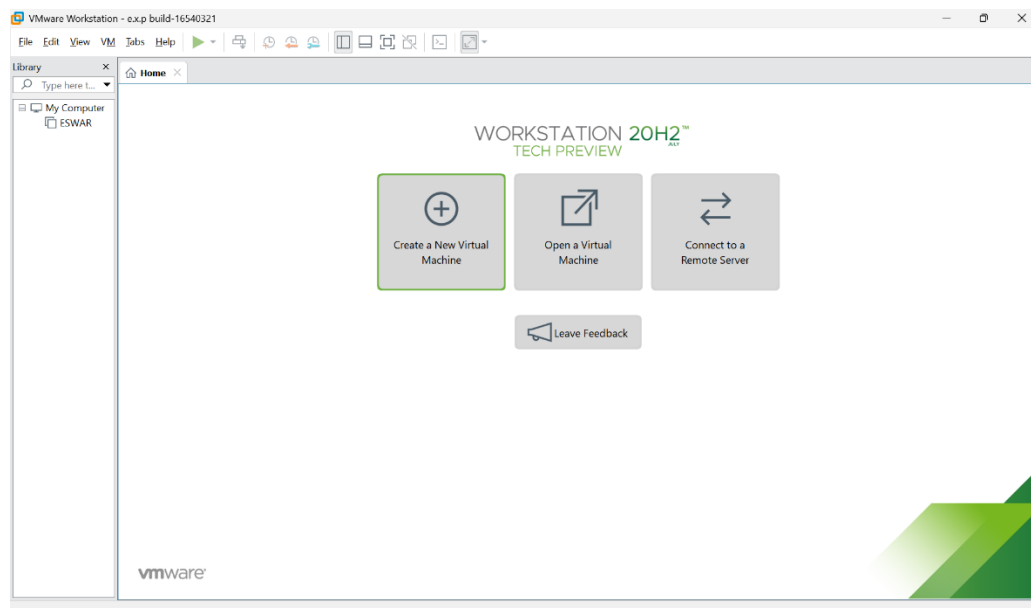
Step 6: Allocate the required CPU, RAM, and storage.

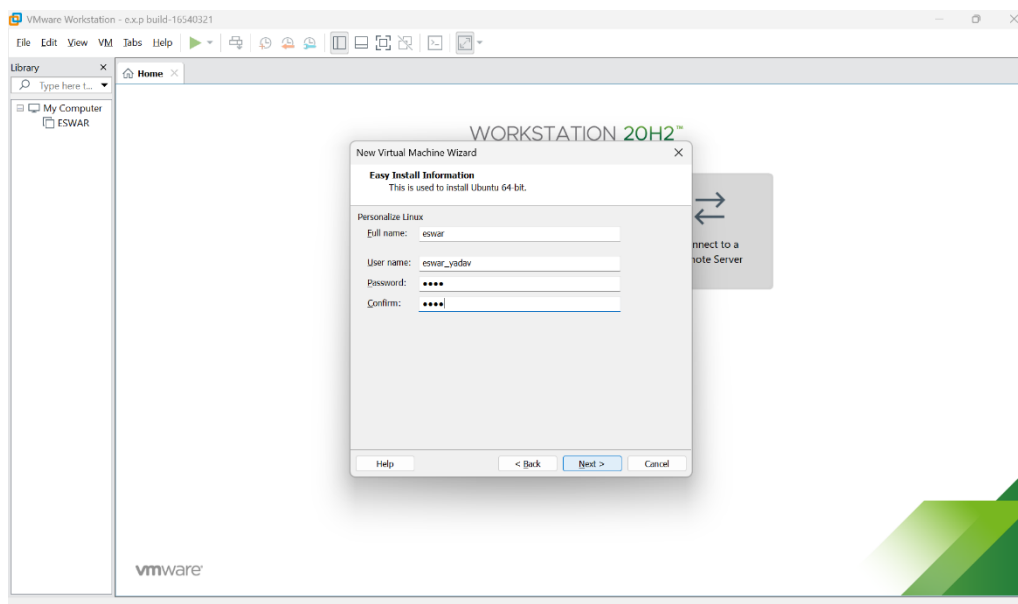
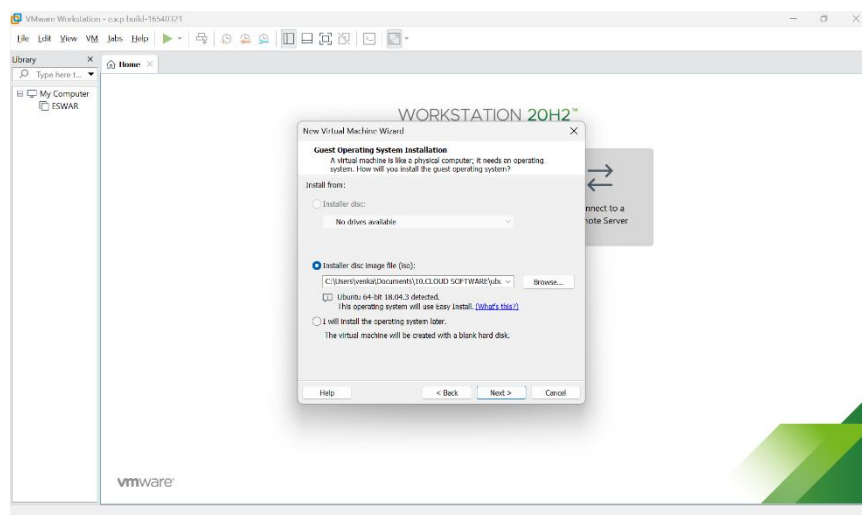
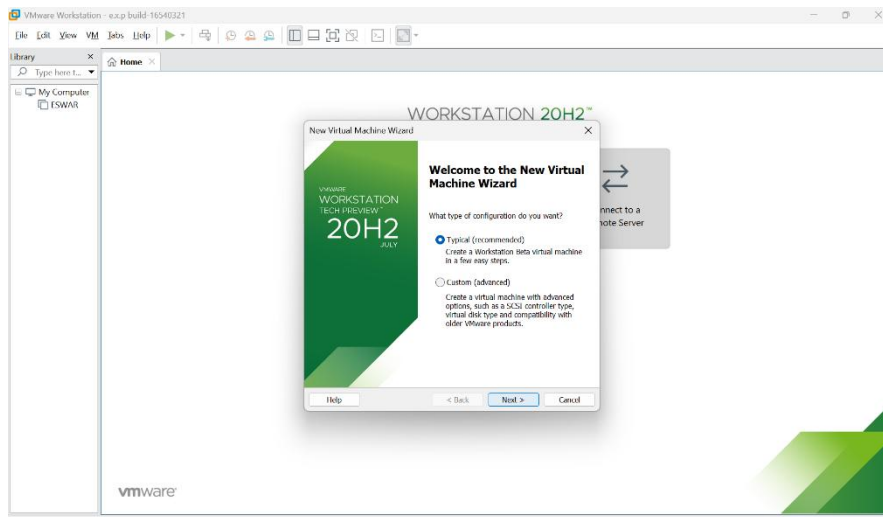
Step 7: Click Finish to create the VM.

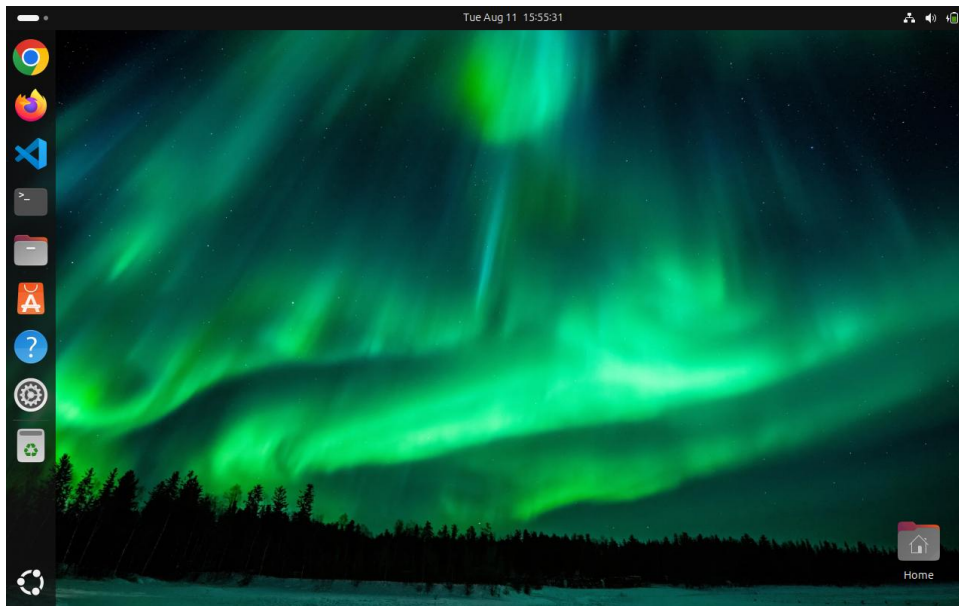
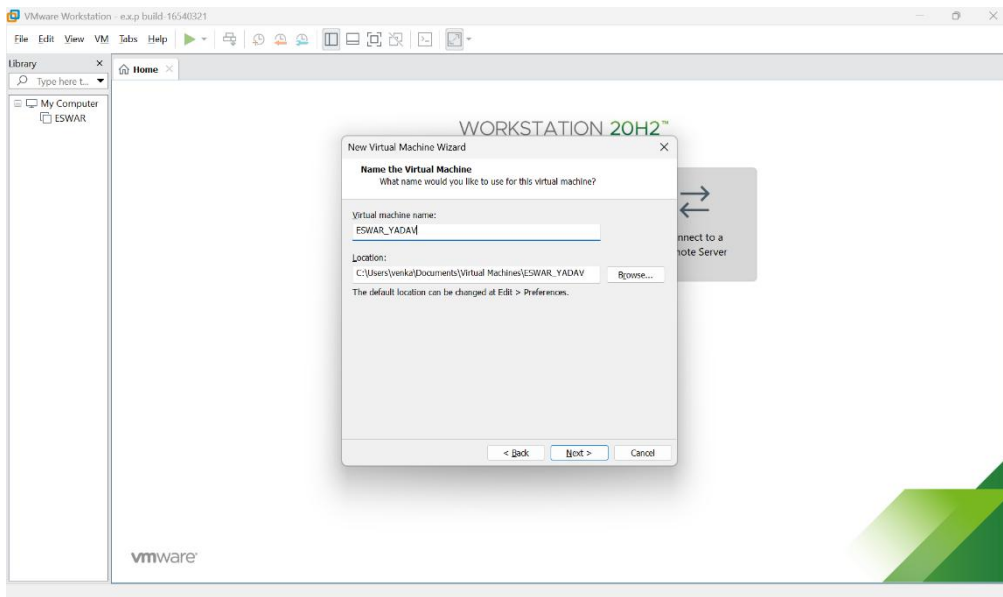
Step 8: Power on the VM and complete the OS installation.

Step 9: Verify that the guest OS is running successfully.

Output







Result

The required **Linux/Windows** operating system was successfully installed and executed as a virtual machine using **VMware Workstation**.

Ex. No: 19 Install a C Compiler and Execute Simple C Programs Using VMware Workstation

Aim

To install a C compiler in a virtual machine and execute simple C programs using VMware Workstation.

PROCEDURE

Step 1: Open VMware Workstation and start the Linux virtual machine.

Step 2: Open the Terminal in the virtual machine.

Step 3: Install the GCC C compiler.

Step 4: Verify the GCC installation using `gcc --version`.

Step 5: Create a simple C program using a text editor.

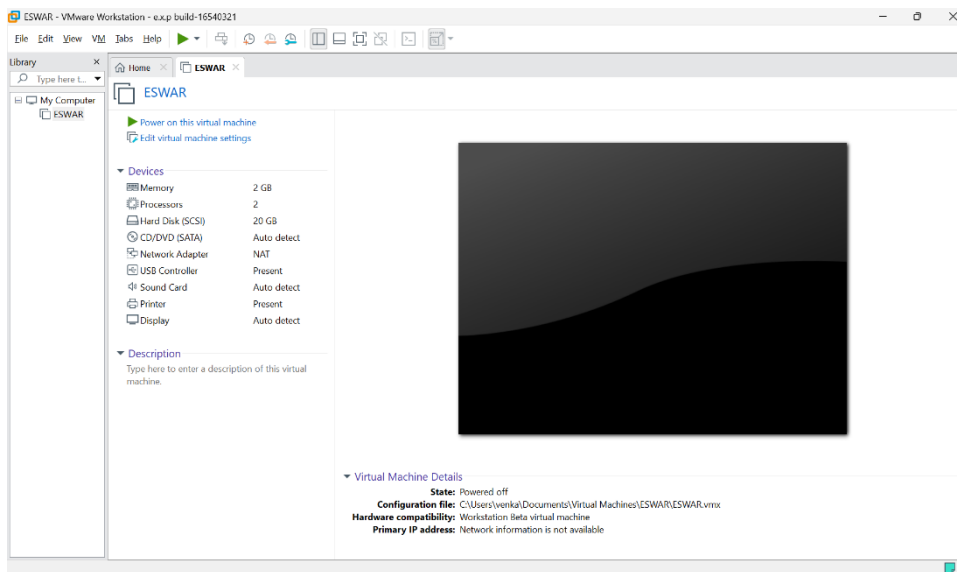
Step 6: Save the program with a `.c` extension.

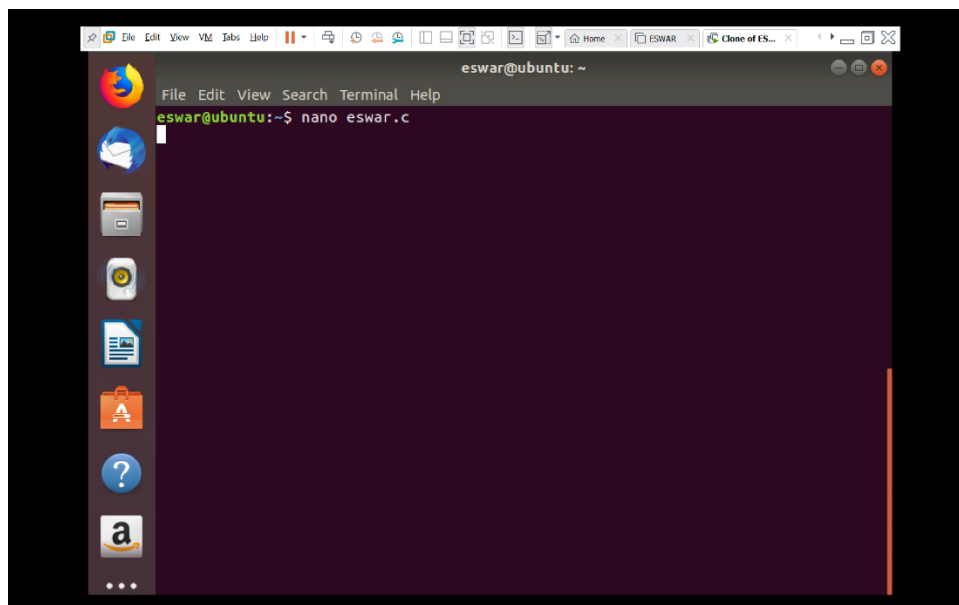
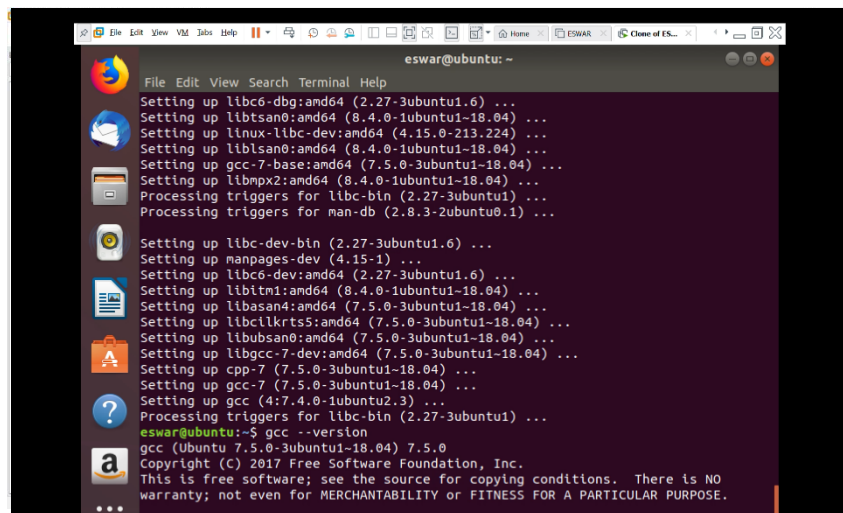
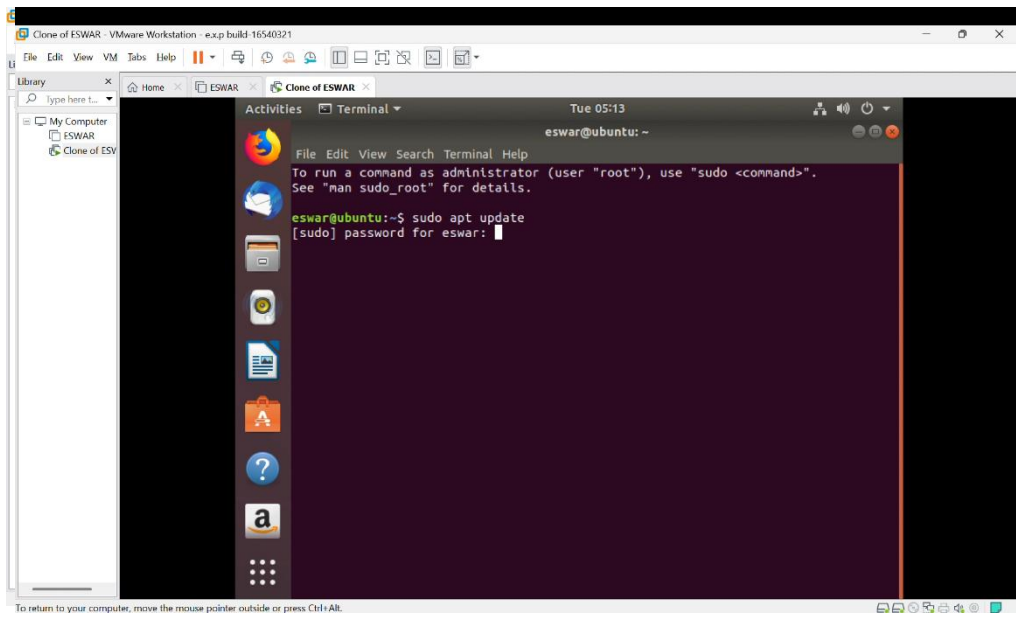
Step 7: Compile the program using the `gcc` command.

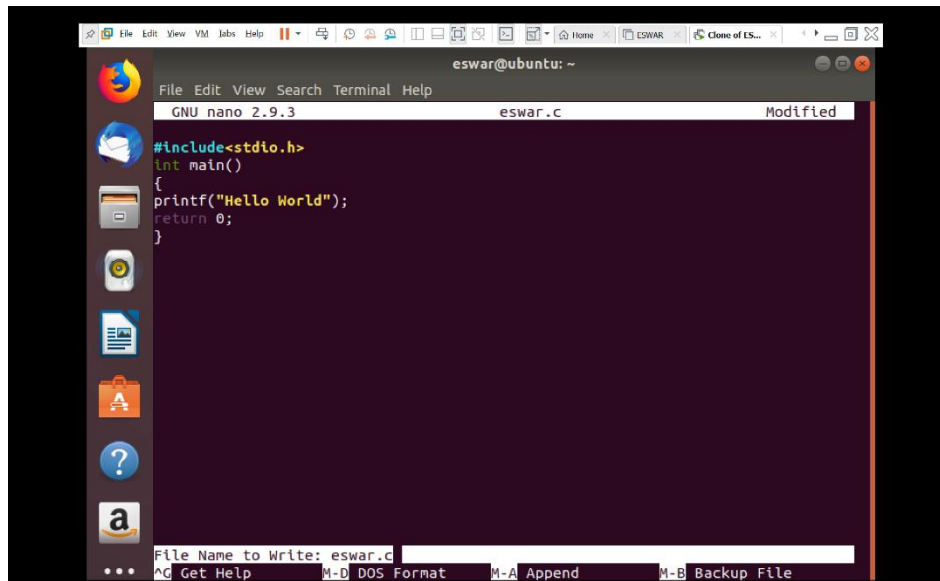
Step 8: Execute the compiled program.

Step 9: Verify the output.

Output



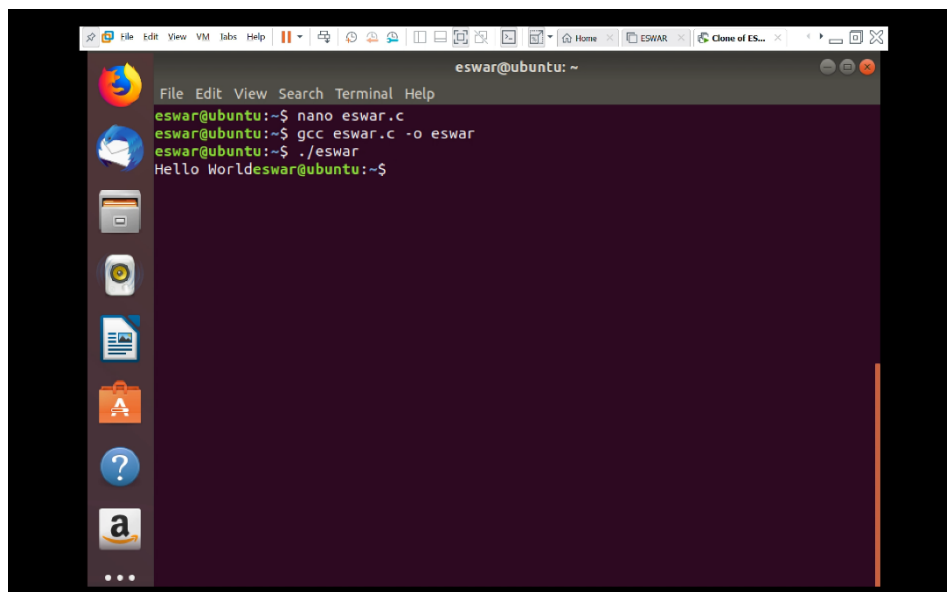




```
eswar@ubuntu: ~
GNU nano 2.9.3 eswar.c Modified

#include<stdio.h>
int main()
{
printf("Hello World");
return 0;
}

File Name to Write: eswar.c
^G Get Help M-D DOS Format M-A Append M-B Backup File
```



```
eswar@ubuntu: ~
File Edit View Search Terminal Help
eswar@ubuntu:~$ nano eswar.c
eswar@ubuntu:~$ gcc eswar.c -o eswar
eswar@ubuntu:~$ ./eswar
Hello World
eswar@ubuntu:~$
```

Result

The **GCC C compiler** was successfully installed, and a simple C program was compiled and executed successfully in the VMware virtual machine.

Ex. No: 20 Install VirtualBox with Different Flavours of Linux/Windows OS Using Custom Installation

Aim

To install VirtualBox and create a virtual machine with a Linux/Windows operating system using custom configuration.

PROCEDURE

PROCEDURE

Step 1: Open Oracle VirtualBox.

Step 2: Click New to create a virtual machine.

Step 3: Enter the VM name and select the required Linux/Windows ISO file.

Step 4: Select the appropriate OS Type and Version.

Step 5: Choose Custom/Manual installation options.

Step 6: Allocate the required RAM and CPU.

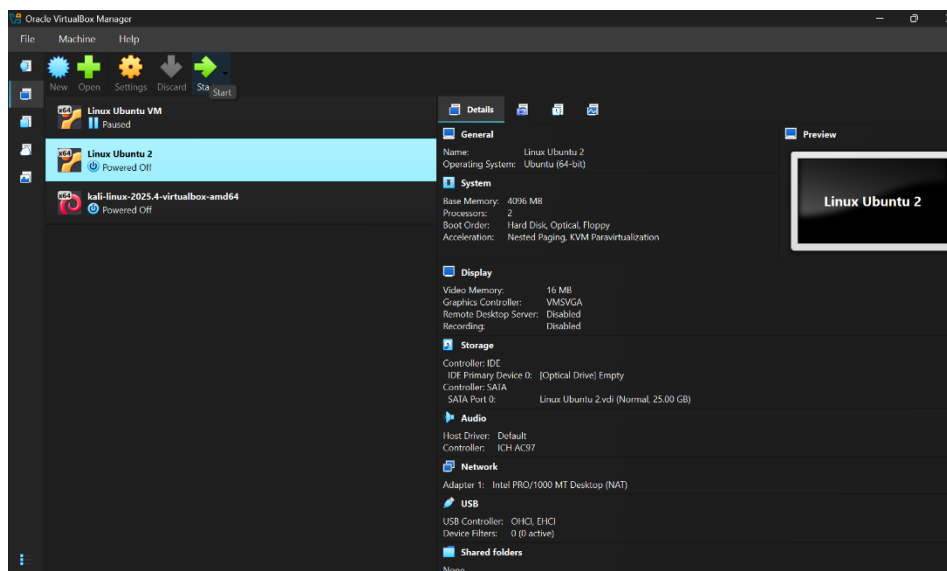
Step 7: Create a virtual hard disk and allocate the required storage.

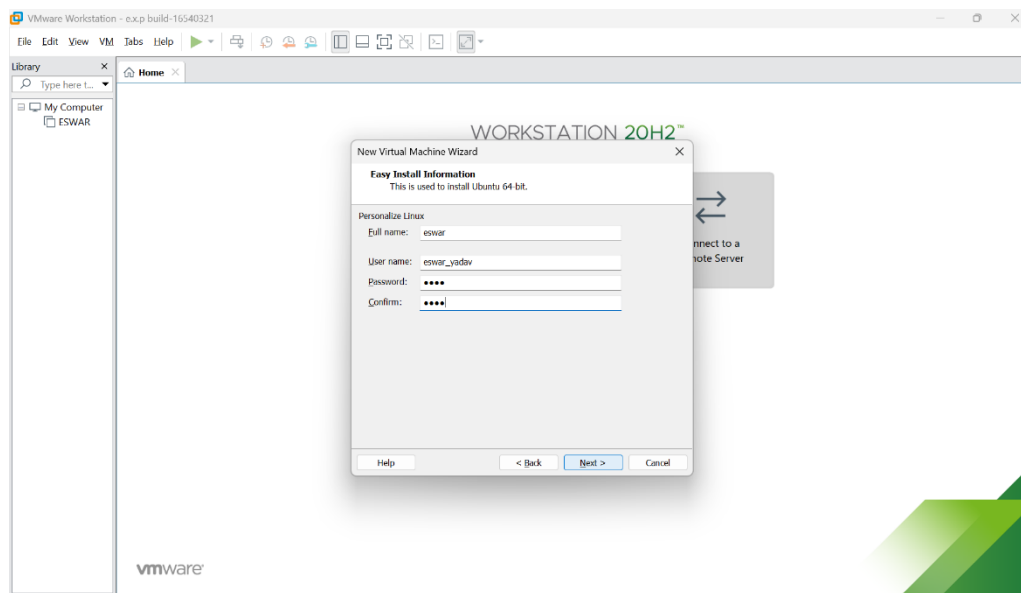
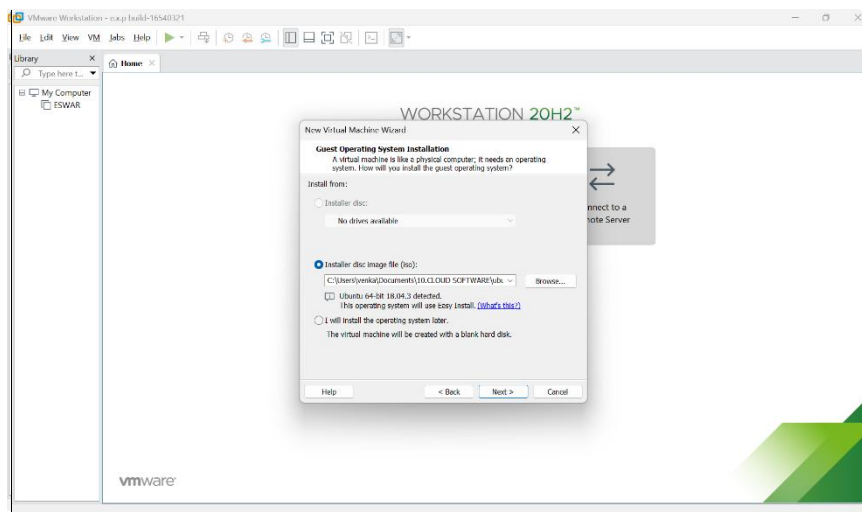
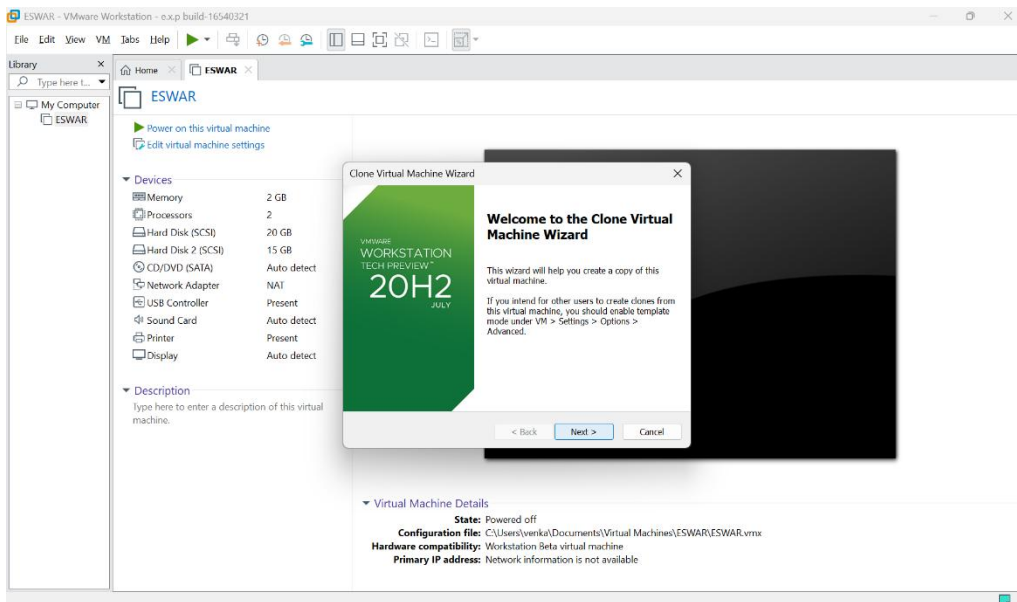
Step 8: Click Finish to create the VM.

Step 9: Start the VM and install the selected operating system.

Step 10: Verify that the installed OS is running successfully in VirtualBox.

Output





Ex. No: 21 Install VMware Workstation with Linux/Windows OS on Windows 7/8

Aim

To install VirtualBox and create a virtual machine with a Linux/Windows operating system using custom configuration.

PROCEDURE

Step 1: Open VMware Workstation on the Windows host.

Step 2: Click Create a New Virtual Machine.

Step 3: Select Typical (Recommended).

Step 4: Select the required Linux/Windows ISO.

Step 5: Enter the VM name and storage location.

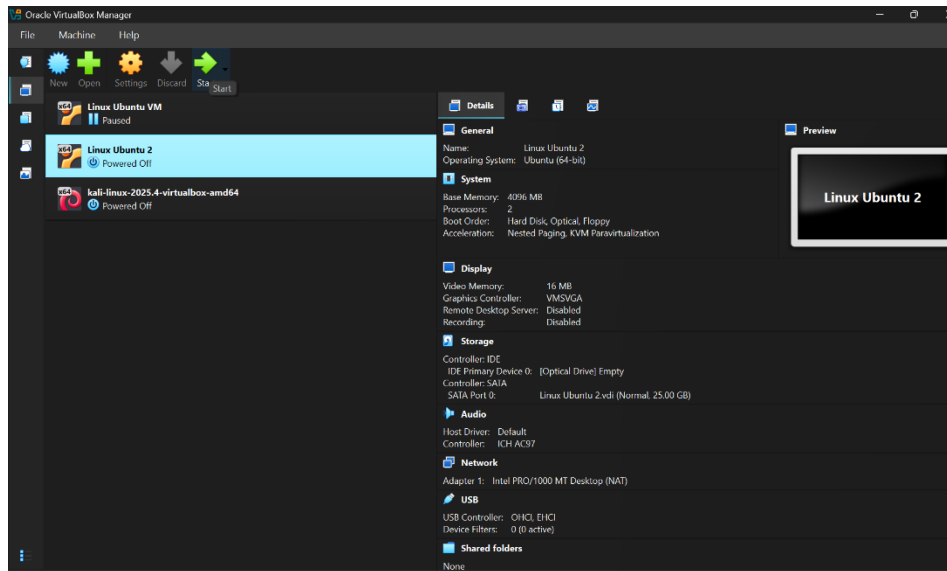
Step 6: Allocate the required CPU, RAM, and storage.

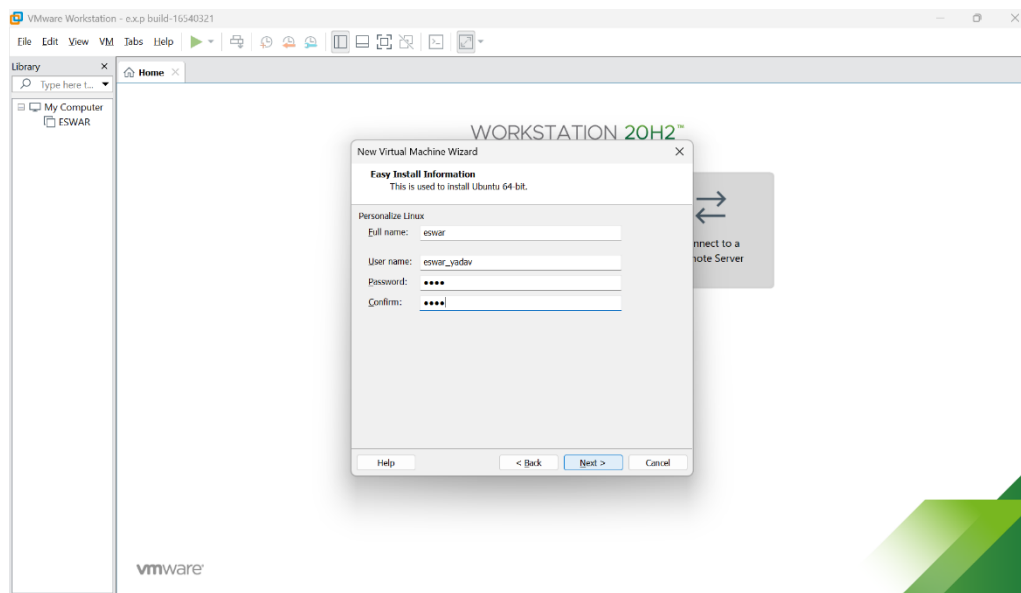
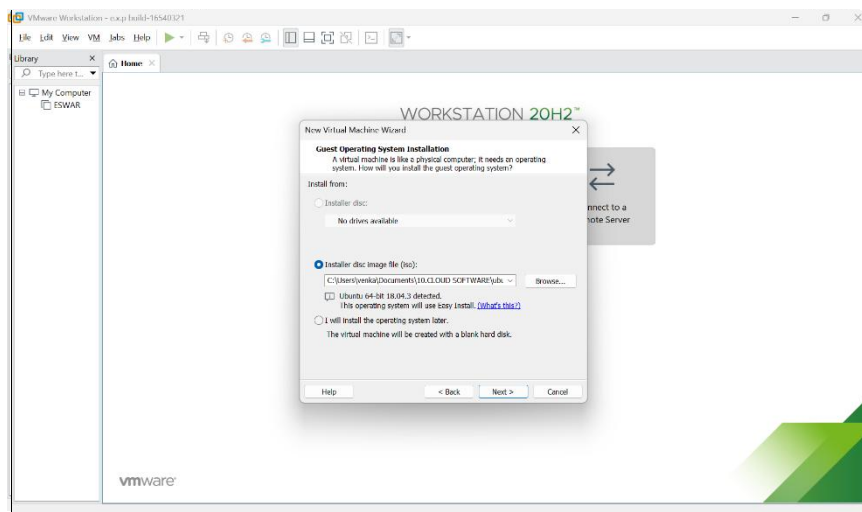
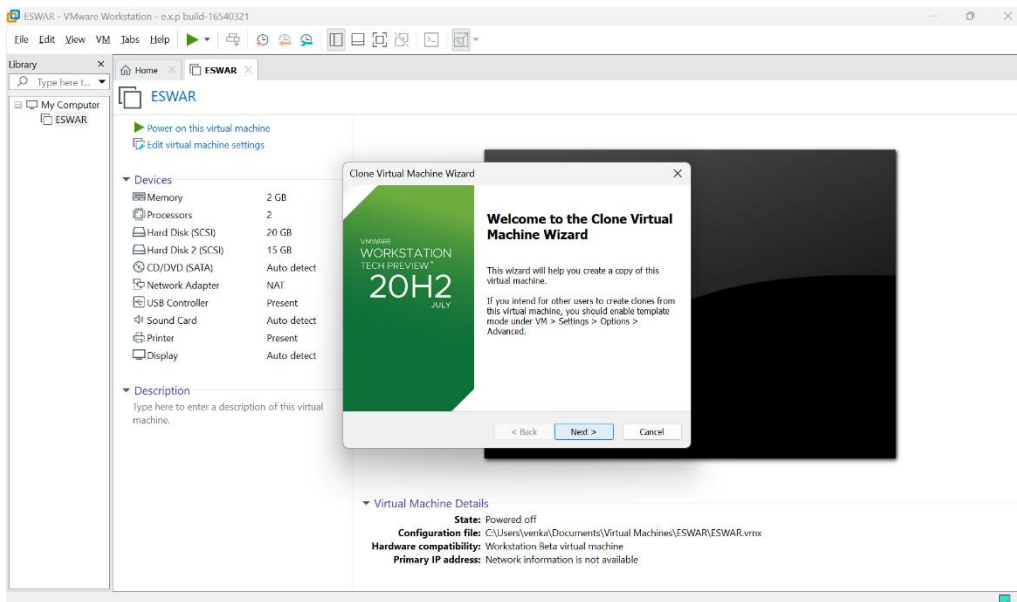
Step 7: Click Finish to create the VM.

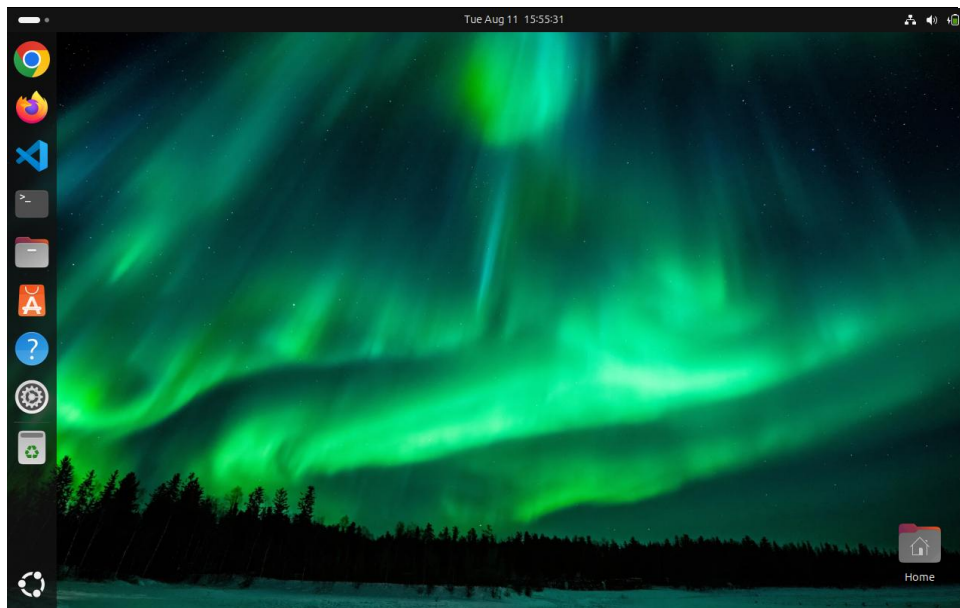
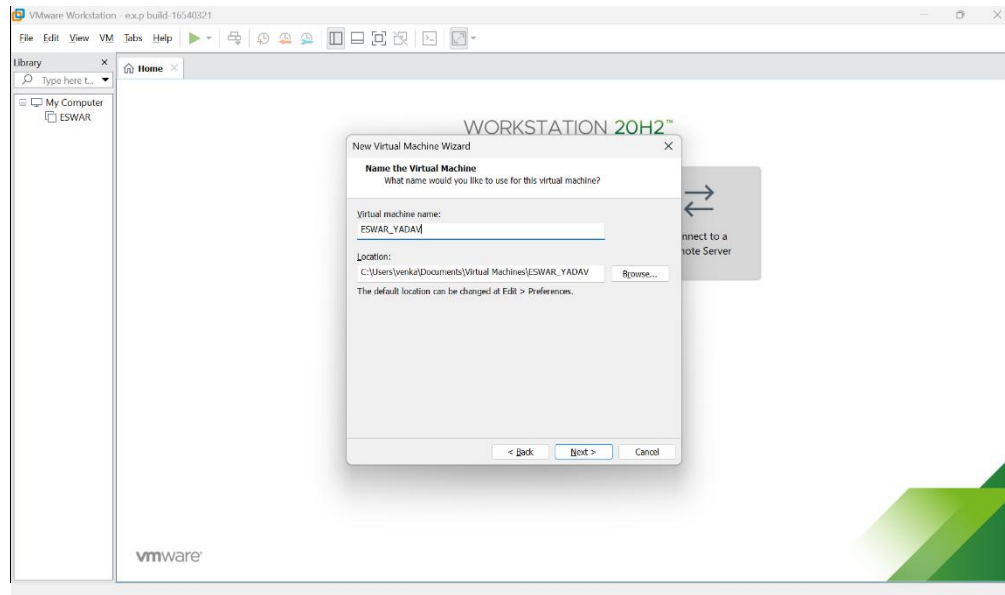
Step 8: Power on the VM and install the selected operating system.

Step 9: Verify that the guest OS is running successfully.

Output







Result

The **Linux/Windows** operating system was successfully installed and executed using **VMware Workstation**.

Ex. No: 22 Run Virtual Machines with Different Configurations and Check the Number of VMs Utilized at a Time

Aim

To install VirtualBox and create a virtual machine with a Linux/Windows operating system using custom configuration.

PROCEDURE

Step 1: Open VMware Workstation.

Step 2: Create or open the required virtual machines.

Step 3: Configure each VM with different CPU, RAM, and storage.

Step 4: Start the first virtual machine and verify that it is working.

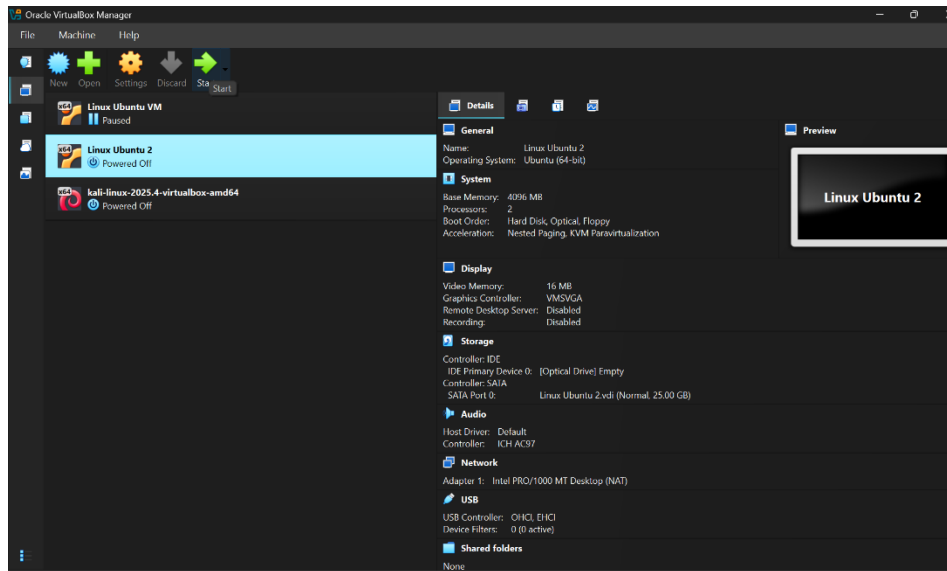
Step 5: Start the second virtual machine.

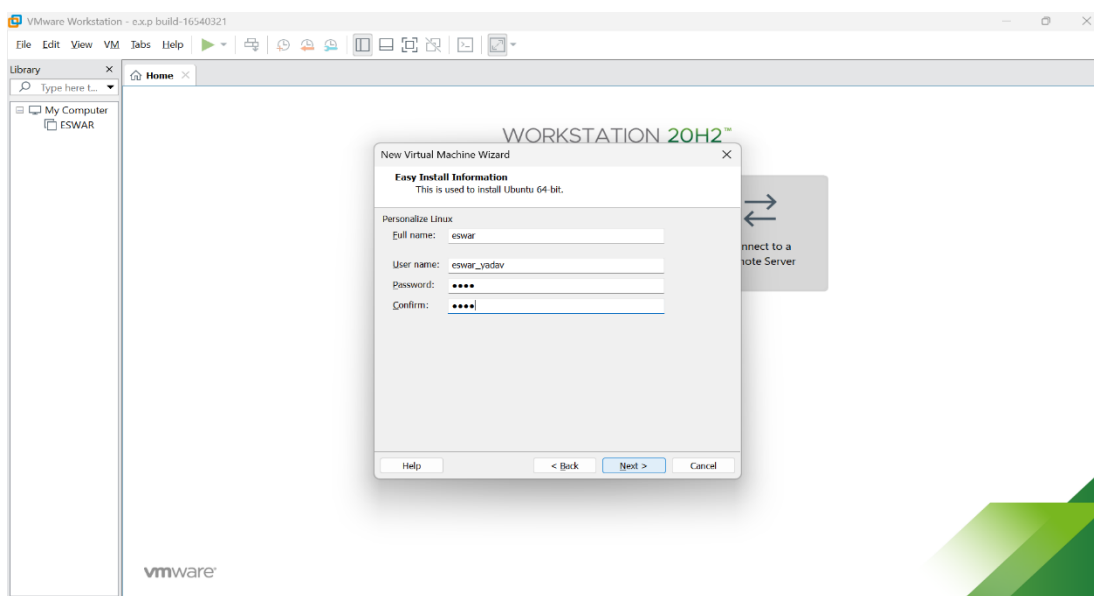
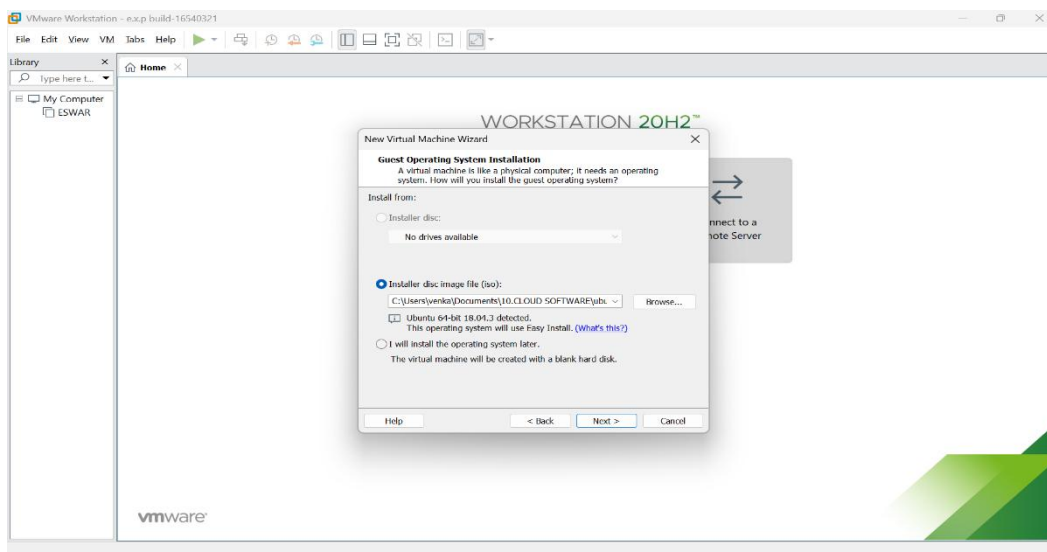
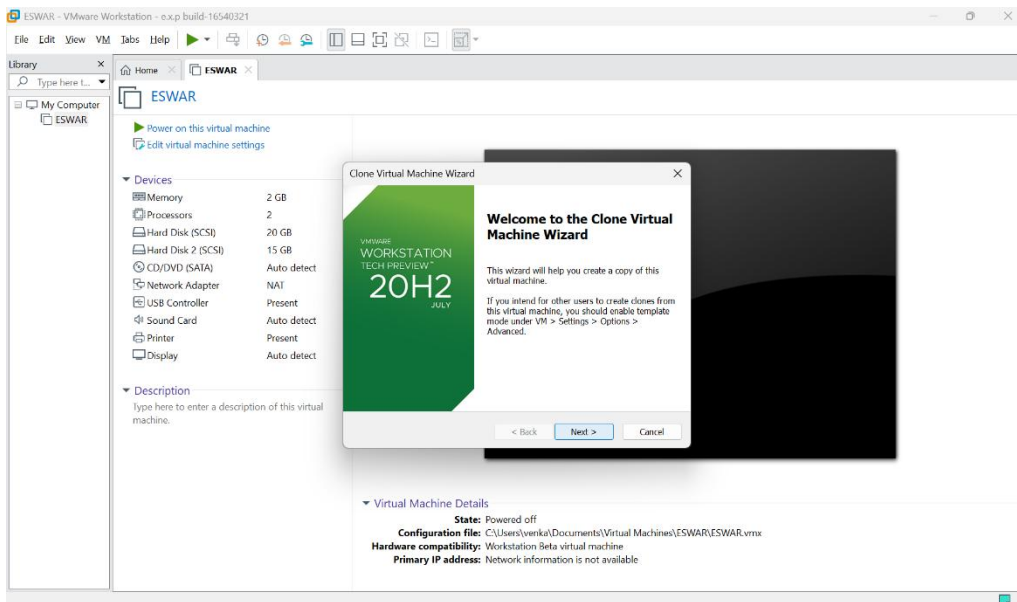
Step 6: Continue starting VMs one by one.

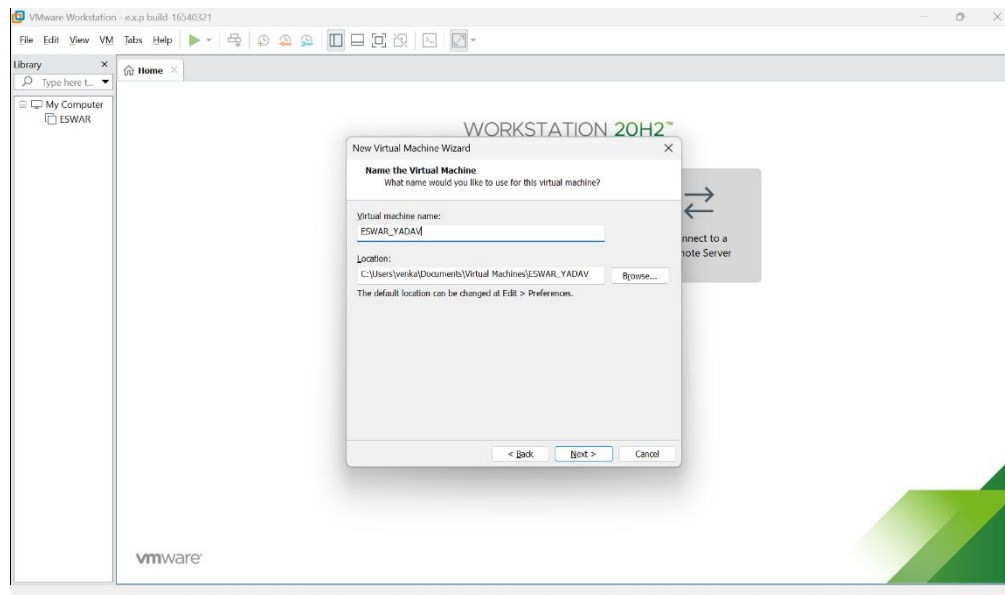
Step 7: Monitor the host computer's CPU and RAM usage.

Step 8: Note the maximum number of VMs that can run smoothly at the same time.

Output







Result

The virtual machines were successfully run with different configurations, and the maximum number of VMs that could operate simultaneously was determined.

Ex. No: 23 Attach a Virtual Block Storage to a Virtual Machine and Verify Data Persistence Using VMware

Workstation

Aim

To attach a virtual storage disk to a virtual machine and verify whether the data remains after releasing the virtual machine.

PROCEDURE

Step 1: Open VMware Workstation and select the virtual machine.

Step 2: Power off the virtual machine.

Step 3: Open VM → Settings → Add.

Step 4: Select Hard Disk and click Next.

Step 5: Select Create a new virtual disk and specify the required storage size.

Step 6: Click Finish and then OK.

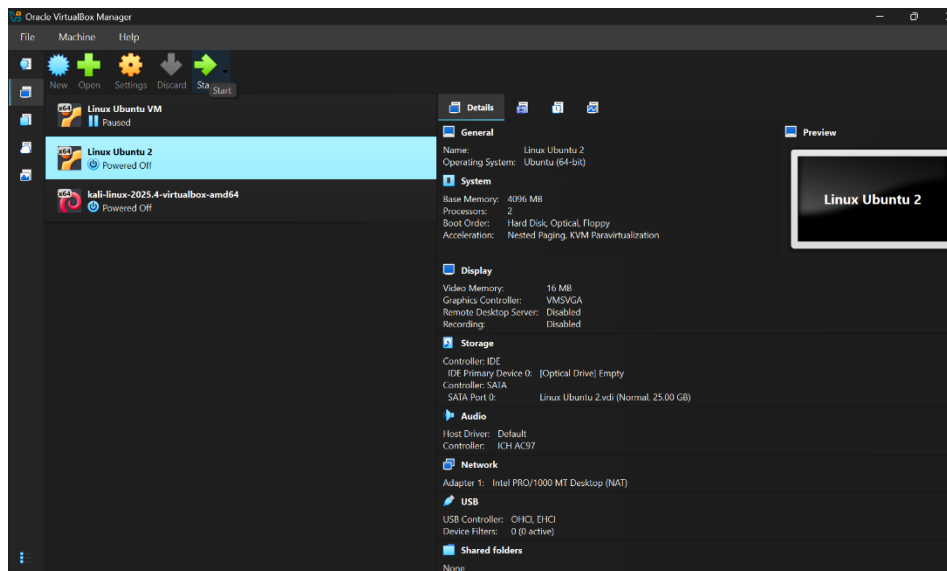
Step 7: Start the virtual machine and open the attached disk.

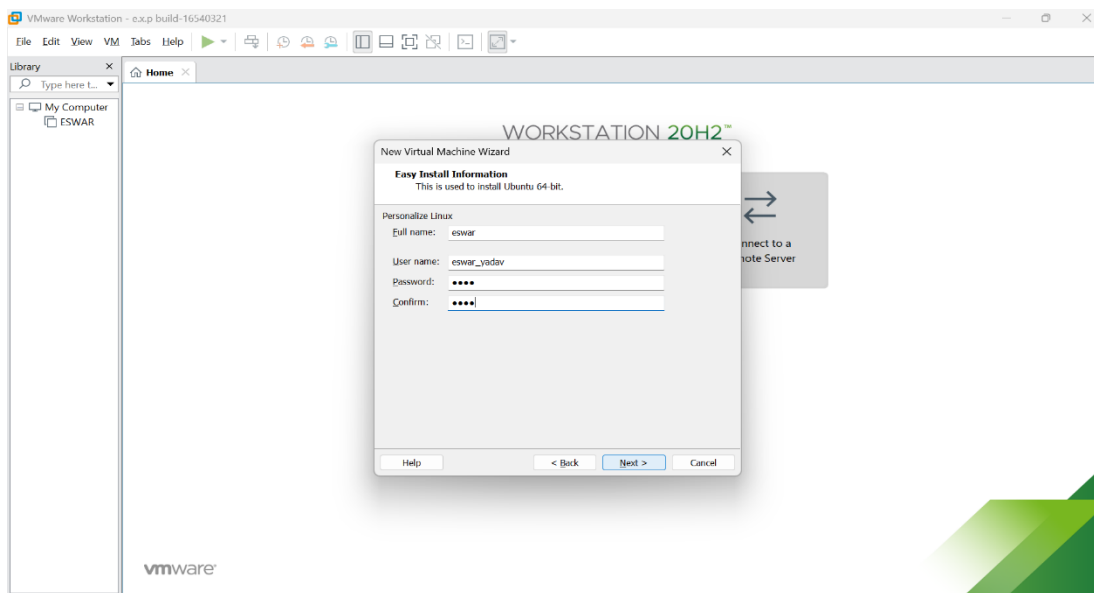
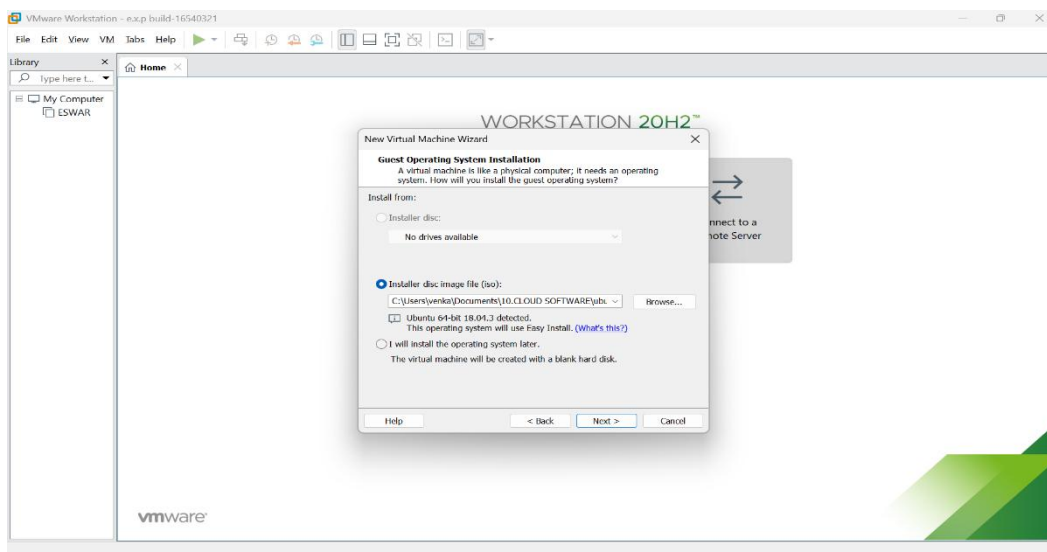
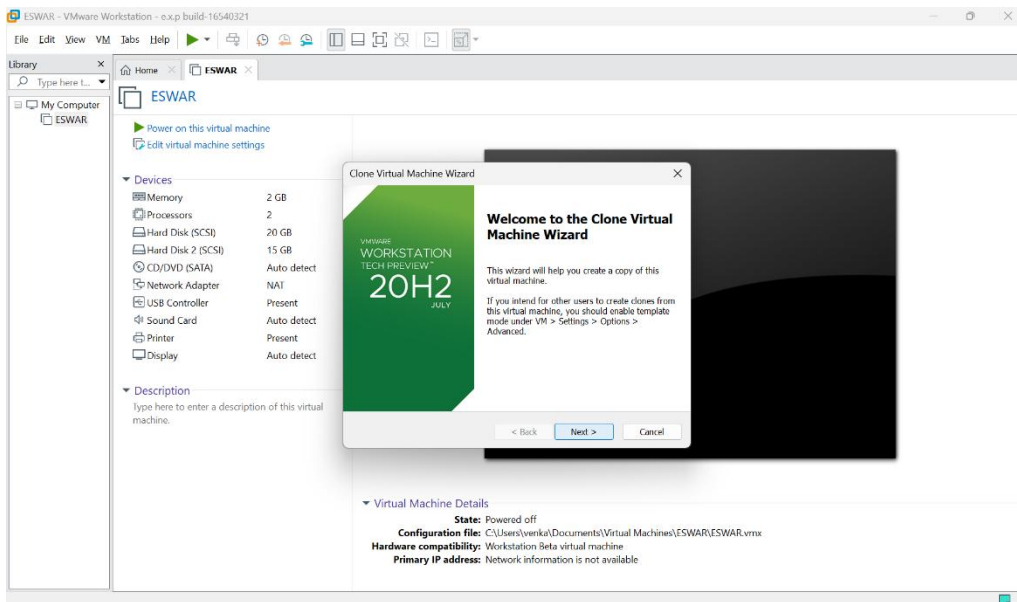
Step 8: Create a sample file and save it on the attached disk.

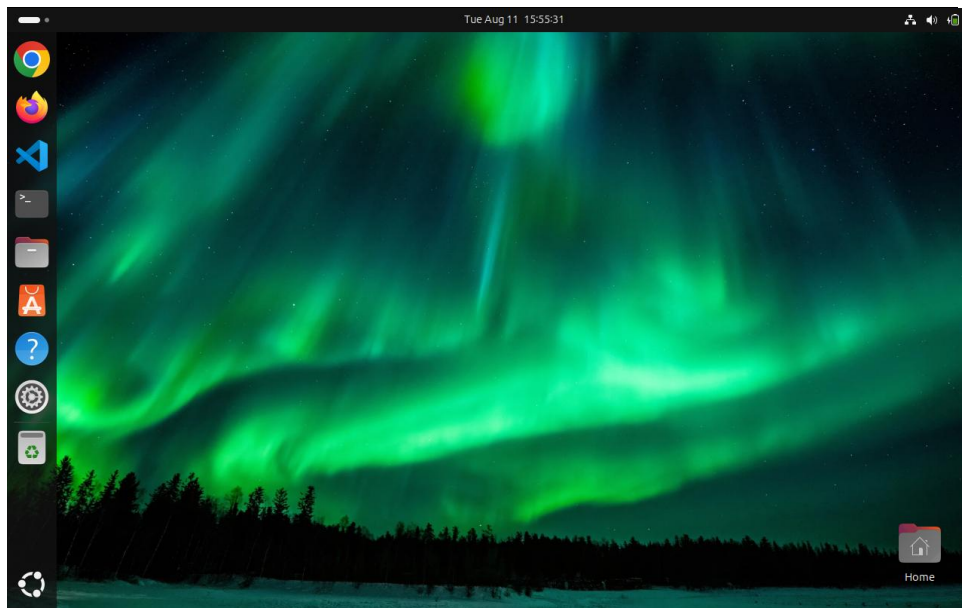
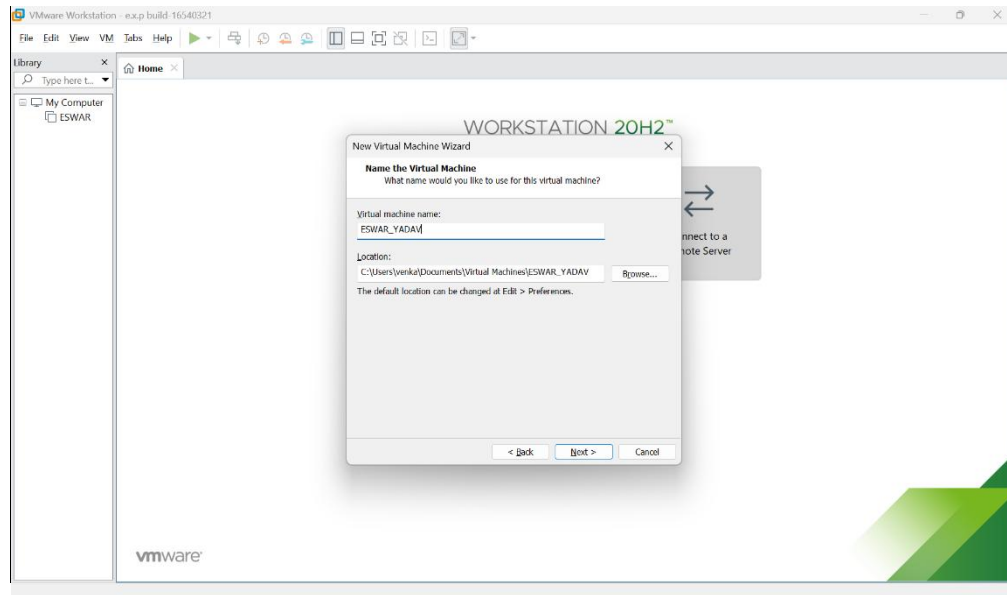
Step 9: Shut down and release/remove the virtual machine.

Step 10: Attach the same virtual disk to another VM and check the saved file.

Output







Result

The virtual block storage was successfully attached, and the stored data remained available after releasing the virtual machine.

Ex. No: 24 Demonstrate Virtual Machine Migration Between Hosts Using VMware

Aim

To demonstrate virtual machine migration from one host to another using VMware.

PROCEDURE

Step 1: Create or select a virtual machine in VMware.

Step 2: Power off the VM and ensure all required VM files are available.

Step 3: Copy the VM files to the second host computer.

Step 4: Open VMware Workstation on the second host.

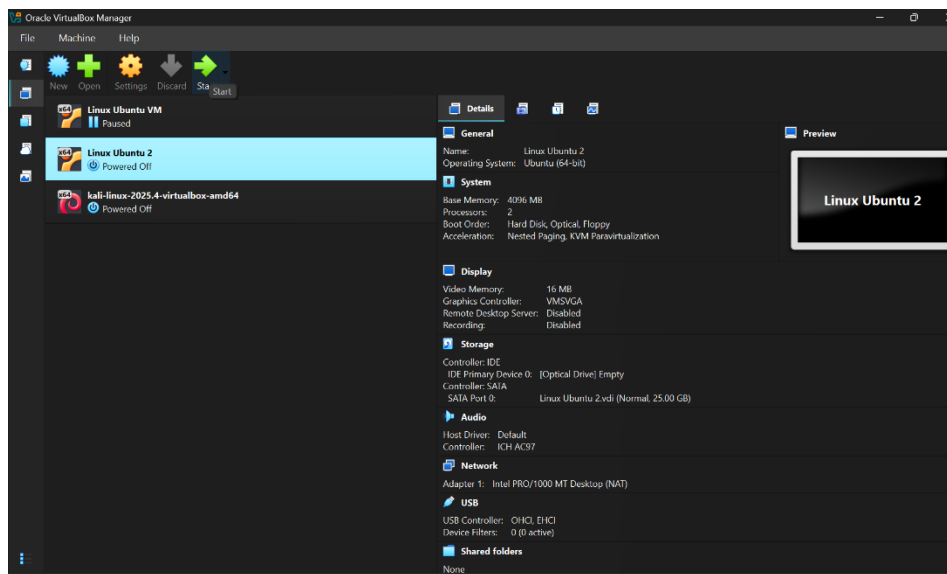
Step 5: Select Open a Virtual Machine.

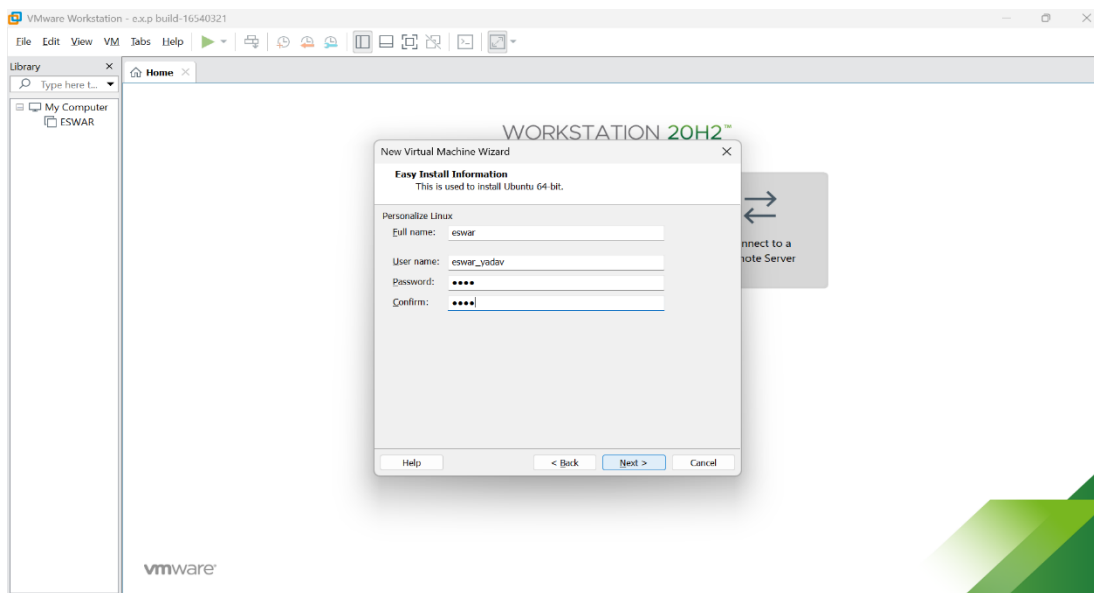
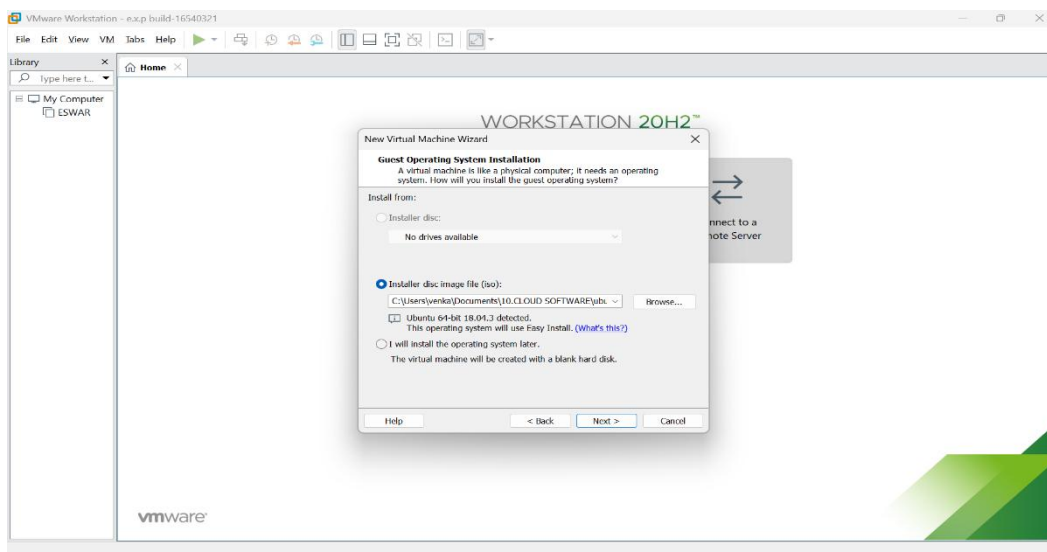
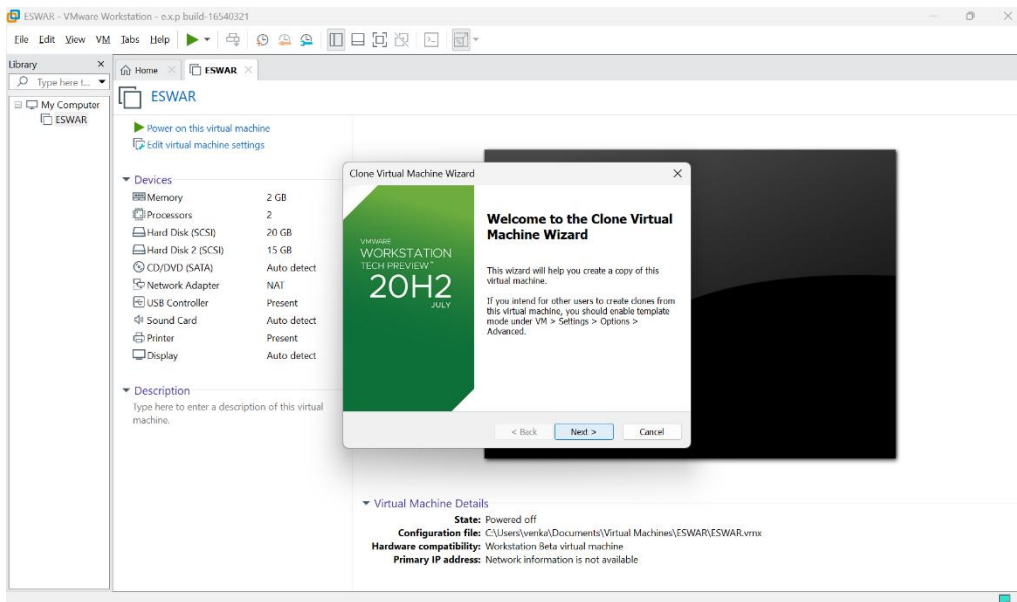
Step 6: Select the copied .vmx file.

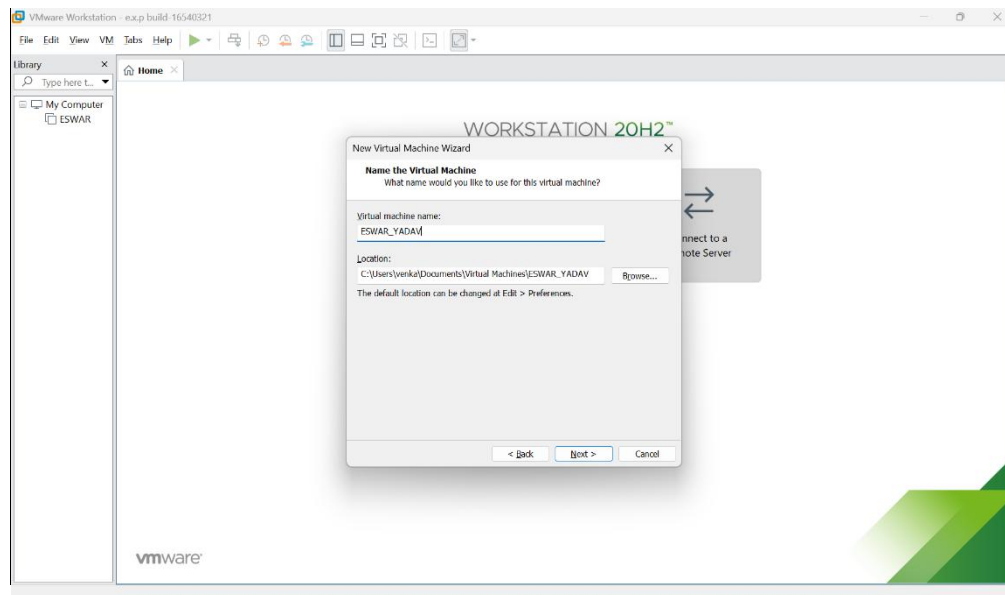
Step 7: Power on the VM on the second host.

Step 8: Verify that the VM and its data are working correctly.

Output







Result

The virtual machine was successfully **migrated from one host to another using VMware**, and its data and configuration were verified.